# A Journey to the World of Numbers

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# The natural numbers

# Starting from 0

The natural numbers form the sequence

$$0, \qquad S(0), \qquad S(S(0)), \qquad S(S(S(0))), \quad \ldots$$

governed by the following principles.

1. $0 \in \mathbb{N}$.
2. If $n \in \mathbb{N}$, then its successor $S(n) \in \mathbb{N}$.
3. Every natural number is either 0 or the successor of a natural number.
4. No successor equals 0: $S(n) \neq 0$.
5. Successors are unique: $S(m) = S(n)$ implies $m = n$.

We set $1 = S(0)$. Once addition is defined, we will prove that $S(n) = n + 1$.

# The induction principle

## Principle of mathematical induction

Let $P(n)$ be a statement about $n \in \mathbb{N}$. If

1. $P(0)$ is true, and
2. for every $n \in \mathbb{N}$, $P(n)$ implies $P(S(n))$,

then $P(n)$ is true for every $n \in \mathbb{N}$.

Induction mirrors the way the natural numbers are generated:

$$\underbrace{P(0)}_{\text{start}} \Longrightarrow P(S(0)) \Longrightarrow P(S(S(0))) \Longrightarrow \cdots$$

# The recursion principle

## Defining a function on $\mathbb{N}$

Let $X$ be a set, choose $x_0 \in X$, and let $G\colon \mathbb{N} \times X \to X$ be a function. There is one and only one function $f\colon \mathbb{N} \to X$ satisfying

$$f(0) = x_0, \qquad f(S(n)) = G(n, f(n)) \quad \text{for every } n \in \mathbb{N}.$$

## Why this matters

Addition, multiplication, and the predecessor function will all be defined by this principle.

# Definition of addition

Fix $a \in \mathbb{N}$. We define $a + b$ by recursion on $b$:

$$\boxed{a + 0 = a}, \qquad \boxed{a + S(b) = S(a + b)}.$$

Thus the second input tells us how many times to take the successor of $a$. For example,

$$\begin{aligned} a + S(S(0)) &= S(a + S(0)) \\ &= S(S(a + 0)) \\ &= S(S(a)). \end{aligned}$$

Since $1 = S(0)$, the defining equations immediately give

$$a + 1 = a + S(0) = S(a + 0) = S(a).$$

# Zero on the left

**Proposition**

For every $a \in \mathbb{N}$, $0 + a = a$.

**Proof.**

We use induction on $a$.

*Base case.* By the first defining equation, $0 + 0 = 0$.

*Inductive step.* Suppose that $0 + n = n$. Then

$$0 + S(n) = S(0 + n) = S(n).$$

The first equality is the second defining equation for addition; the second uses the induction hypothesis. Hence the statement passes from $n$ to $S(n)$.

By induction, $0 + a = a$ for all $a \in \mathbb{N}$. $\square$

# Moving a successor through a sum

## Proposition

For all $a, b \in \mathbb{N}$,

$$S(a) + b = S(a + b).$$

## Proof.

Fix $a$ and use induction on $b$.

*Base case.* $S(a) + 0 = S(a) = S(a + 0)$.

*Inductive step.* Assume $S(a) + n = S(a + n)$. Then

$$\begin{aligned} S(a) + S(n) &= S(S(a) + n) && \text{by the definition of addition} \\ &= S(S(a + n)) && \text{by the induction hypothesis} \\ &= S(a + S(n)) && \text{by the definition of addition.} \end{aligned}$$

This proves the required identity for $S(n)$. □

# Associativity of addition

## Theorem

For all $a, b, c \in \mathbb{N}$,

$$(a+b)+c = a+(b+c).$$

## Proof.

Fix $a, b$ and use induction on $c$.
*Base case.* $(a+b)+0 = a+b = a+(b+0)$.
*Inductive step.* Suppose $(a+b)+n = a+(b+n)$. Then

$$\begin{aligned}(a+b)+S(n) &= S((a+b)+n)\\ &= S(a+(b+n))\\ &= a+S(b+n)\\ &= a+(b+S(n)).\end{aligned}$$

Here the second equality is the induction hypothesis, and the other equalities are the defining equation for addition, read in either direction. $\square$

# Commutativity of addition

## Theorem

For all $a, b \in \mathbb{N}$, $a + b = b + a$.

## Proof.

Fix $b$ and use induction on $a$.

*Base case.* We already proved $0 + b = b$, while $b + 0 = b$ by definition. Hence $0 + b = b + 0$.

*Inductive step.* Suppose $n + b = b + n$. Using the formula for moving a successor through a sum,

$$\begin{aligned} S(n) + b &= S(n + b) && \\ &= S(b + n) && \text{by the induction hypothesis} \\ &= b + S(n) && \text{by the definition of addition.} \end{aligned}$$

Therefore the equality holds for $S(n)$, and induction completes the proof. $\square$

# When can a sum be zero?

## Proposition

If $a + b = 0$, then $a = 0$ and $b = 0$.

## Proof.

First inspect $b$.
If $b = 0$, then $a + b = a + 0 = a$, so the assumption gives $a = 0$.
If $b = S(n)$, then

$$a + b = a + S(n) = S(a + n),$$

which cannot equal 0. Thus the second possibility is ruled out, and $b = 0$; the preceding paragraph then gives $a = 0$. $\square$

Conversely, $0 + 0 = 0$, so in fact

$$a + b = 0 \iff a = 0 \text{ and } b = 0.$$

# Cancellation for addition

## Theorem

If $a + b = a + c$, then $b = c$.

## Proof.

We use induction on $a$, proving the assertion simultaneously for all $b, c \in \mathbb{N}$.
*Base case.* If $0 + b = 0 + c$, then $b = c$, because $0 + b = b$ and $0 + c = c$.
*Inductive step.* Assume cancellation is valid with $n$ on the left. If

$$S(n) + b = S(n) + c,$$

moving the successors through the sums gives $S(n + b) = S(n + c)$. Uniqueness of successors yields $n + b = n + c$, and the induction hypothesis now yields $b = c$. $\square$

# Definition of multiplication

Fix $a \in \mathbb{N}$. We define $ab$ by recursion on $b$:

$$\boxed{a \cdot 0 = 0}, \qquad \boxed{a \cdot S(b) = ab + a}.$$

Multiplication is therefore repeated addition. For example,

$$\begin{aligned} a \cdot S(S(0)) &= a \cdot S(0) + a \\ &= (a \cdot 0 + a) + a \\ &= (0 + a) + a \\ &= a + a. \end{aligned}$$

# Moving a successor through a product

## Proposition

For all $a, b \in \mathbb{N}$,

$$S(a) \cdot b = ab + b.$$

## Proof.

Fix $a$ and use induction on $b$.
*Base case.* $S(a) \cdot 0 = 0 = a \cdot 0 + 0$.
*Inductive step.* Suppose $S(a) \cdot n = an + n$. Then

$$\begin{aligned} S(a) \cdot S(n) &= S(a) \cdot n + S(a) \\ &= (an + n) + S(a) \\ &= S((an + n) + a) \\ &= S((an + a) + n) \\ &= a \cdot S(n) + S(n). \end{aligned}$$

Associativity and commutativity of addition justify the fourth equality. □

# Zero on the left of a product

## Proposition

For every $a \in \mathbb{N}$, $0 \cdot a = 0$.

## Proof.

We use induction on $a$.
*Base case.* $0 \cdot 0 = 0$ by definition.
*Inductive step.* Suppose $0 \cdot n = 0$. Then

$$0 \cdot S(n) = 0 \cdot n + 0 = 0 + 0 = 0.$$

Thus the property passes to the successor, and induction proves it for every $a$. □

Together with the defining identity $a \cdot 0 = 0$, this says that 0 is absorbing on both sides of multiplication.

# Left distributivity

## Theorem

For all $a, b, c \in \mathbb{N}$,
$$a(b + c) = ab + ac.$$

## Proof.

Fix $a, b$ and use induction on $c$.
*Base case.* $a(b + 0) = ab = ab + 0 = ab + a \cdot 0$.
*Inductive step.* Suppose $a(b + n) = ab + an$. Then

$$\begin{aligned} a(b + S(n)) &= aS((b + n)) \\ &= a(b + n) + a \\ &= (ab + an) + a \\ &= ab + (an + a) \\ &= ab + aS(n). \end{aligned}$$

We used the recursion equations and associativity of addition. □

# The multiplicative identity

## Proposition

For every $a \in \mathbb{N}$, $a \cdot 1 = a$.

Since $1 = S(0)$, the definition of multiplication gives

$$a \cdot 1 = a \cdot S(0) = a \cdot 0 + a = 0 + a = a.$$

## A useful observation

The equality $1 \cdot a = a$ will follow once multiplication is known to be commutative. It can also be proved directly by induction on $a$.

The base case is $1 \cdot 0 = 0$. If $1 \cdot n = n$, then

$$1 \cdot S(n) = 1 \cdot n + 1 = n + 1 = S(n).$$

# Commutativity of multiplication

## Theorem

For all $a, b \in \mathbb{N}$, $ab = ba$.

## Proof.

Fix $a$ and use induction on $b$.

*Base case.* $a \cdot 0 = 0 = 0 \cdot a$.

*Inductive step.* Suppose $an = na$. Then

$$\begin{aligned} a \cdot S(n) &= an + a && \text{by the definition of multiplication} \\ &= na + a && \text{by the induction hypothesis} \\ &= S(n) \cdot a && \text{by the successor–product formula.} \end{aligned}$$

Hence $a \cdot b = b \cdot a$ for all $b$. $\square$

# Right distributivity

## Theorem

For all $a, b, c \in \mathbb{N}$,

$$(a + b)c = ac + bc.$$

## Proof.

We convert right multiplication into left multiplication, use left distributivity, and convert back:

$$\begin{aligned} (a+b)c &= c(a+b) && \text{commutativity of multiplication} \\ &= ca + cb && \text{left distributivity} \\ &= ac + bc && \text{commutativity of multiplication.} \end{aligned}$$

$\square$

This short argument illustrates an important strategy: a symmetry theorem can turn a new statement into one already proved.

# Associativity of multiplication

## Theorem

For all $a, b, c \in \mathbb{N}$,

$$(ab)c = a(bc).$$

## Proof.

Fix $a, b$ and use induction on $c$.
*Base case.* $(ab)0 = 0 = a0 = a(b0)$.
*Inductive step.* Suppose $(ab)n = a(bn)$. Then

$$\begin{aligned}(ab)S(n) &= (ab)n + ab \\ &= a(bn) + ab \\ &= a(bn + b) \\ &= a(bS(n)).\end{aligned}$$

The third equality uses left distributivity in reverse. □

# No zero divisors

## Theorem

If $a \neq 0$ and $b \neq 0$, then $ab \neq 0$.

Since every nonzero natural number is a successor, write $a = S(p) = p + 1$ and $b = S(q) = q + 1$. Then

$$\begin{aligned} ab &= (p+1)(q+1) \\ &= p(q+1) + 1(q+1) \\ &= (pq+p) + (q+1) \\ &= (pq+p+q) + 1 \\ &= S(pq+p+q). \end{aligned}$$

The final expression is a successor, so it cannot be 0. Therefore $ab \neq 0$.

## Equivalent form

If $ab = 0$, then $a = 0$ or $b = 0$.

# The predecessor function

Define $\operatorname{pred}\colon \mathbb{N} \to \mathbb{N}$ by

$$\operatorname{pred}(0) = 0, \qquad \operatorname{pred}(S(a)) = a.$$

Consequently,

$$\operatorname{pred}(a+1) = a$$

for every $a$, because $a + 1 = S(a)$.
Conversely, if $a \neq 0$, then $a$ is a successor, say $a = S(n)$, and

$$\operatorname{pred}(a) + 1 = \operatorname{pred}(S(n)) + 1 = n + 1 = S(n) = a.$$

The condition $a \neq 0$ is essential: our definition gives $\operatorname{pred}(0) + 1 = 1 \neq 0$.

# The algebra obtained so far

For all $a, b, c \in \mathbb{N}$, we have established

$$\begin{aligned} a + b &= b + a, & (a+b)+c &= a+(b+c), & a + 0 &= a, \\ ab &= ba, & (ab)c &= a(bc), & a \cdot 1 &= a, \\ a(b+c) &= ab + ac, & (a+b)c &= ac + bc, & a \cdot 0 &= 0. \end{aligned}$$

Thus $\mathbb{N}$, together with $0, 1, +, \cdot$, is a commutative semiring. In addition,

- $0 \neq 1$;
- addition admits cancellation;
- a product of two nonzero numbers is nonzero.

We now build the order directly from addition.

# Definition of $\leq$

## Definition

For $a, b \in \mathbb{N}$, we write

$$\boxed{a \leq b} \quad \text{if and only if} \quad \boxed{\text{there is an } x \in \mathbb{N} \text{ such that } b = a + x.}$$

The number $x$ is the gap from $a$ to $b$.

Immediate examples are

- $0 \leq a$, because $a = 0 + a$;
- $a \leq a$, because $a = a + 0$;
- $a \leq S(a)$, because $S(a) = a + 1$;
- $a \leq a + b$, because $a + b = a + b$.

We must prove that this relation really has all the expected order properties.

# Reflexivity of $\leq$

## Proposition

For every $a \in \mathbb{N}$,

$$a \leq a.$$

## Proof.

By definition, we must find an $x \in \mathbb{N}$ such that

$$a = a + x.$$

Choose $x = 0$. The additive identity gives

$$a = a + 0,$$

so 0 is a gap from $a$ to itself. Hence $a \leq a$. $\square$

# Transitivity of $\leq$

## Proposition

If $a \leq b$ and $b \leq c$, then $a \leq c$.

## Proof.

Since $a \leq b$, choose $x \in \mathbb{N}$ such that

$$b = a + x.$$

Since $b \leq c$, choose $y \in \mathbb{N}$ such that

$$c = b + y.$$

Substituting the first equality into the second and using associativity,

$$c = (a + x) + y = a + (x + y).$$

Because $x + y \in \mathbb{N}$, it is a gap from $a$ to $c$. Hence $a \leq c$. $\square$

# Antisymmetry of $\leq$: setting up the gaps

**Theorem**

If $a \leq b$ and $b \leq a$, then $a = b$.

Choose gaps $x, y \in \mathbb{N}$ such that

$$b = a + x, \qquad a = b + y.$$

Substitution gives

$$a = b + y = (a + x) + y = a + (x + y).$$

But also $a = a + 0$. Therefore

$$a + 0 = a + (x + y).$$

Cancellation for addition yields

$$0 = x + y.$$

# Antisymmetry of $\leq$: finishing the proof

From $x + y = 0$, the zero-sum proposition gives $x = 0$ and $y = 0$. In particular, the first gap vanishes, and therefore

$$b = a + x = a + 0 = a.$$

Hence $a = b$, as required.

Notice the chain of earlier results used here:

$$\text{associativity} \implies \text{addition cancellation} \implies \text{antisymmetry}.$$

# Totality of $\leq$: the induction

**Theorem**

For every $a, b \in \mathbb{N}$, either $a \leq b$ or $b \leq a$.

Fix $b$ and use induction on $a$.
*Base case.* For $a = 0$, we have $0 \leq b$, witnessed by the gap $b$.
*Inductive step.* Suppose that either $n \leq b$ or $b \leq n$. We must compare $S(n)$ with $b$.
If $b \leq n$, then $b \leq n \leq S(n)$, so transitivity gives $b \leq S(n)$.
It remains to study the case $n \leq b$.

# Totality of $\leq$: analysing the remaining gap

Suppose $n \leq b$. Choose $x \in \mathbb{N}$ with $b = n + x$. There are two possibilities.

- If $x = 0$, then $b = n$, and hence $b = n \leq S(n)$.
- If $x = S(y)$, then

$$\begin{aligned} b = n + S(y) &= S(n + y) \\ &= S(n) + y. \end{aligned}$$

  Thus $y$ is a gap from $S(n)$ to $b$, so $S(n) \leq b$.

In every case, $S(n) \leq b$ or $b \leq S(n)$. The inductive step is complete, and therefore the order is total.

# A useful successor test

## Proposition

For all $a, b \in \mathbb{N}$,

$$a \leq S(b) \iff a = S(b) \text{ or } a \leq b.$$

Suppose first that $S(b) = a + x$.

- If $x = 0$, then $S(b) = a$, so $a = S(b)$.
- If $x = S(y)$, then $S(b) = a + S(y) = S(a + y)$. Uniqueness of successors gives $b = a + y$, hence $a \leq b$.

Conversely, equality gives $a \leq S(b)$ by reflexivity; and if $a \leq b$, then $a \leq b \leq S(b)$ by transitivity.

# We have a linear order

The relation $\leq$ is

- reflexive: $a \leq a$;
- transitive: $a \leq b \leq c$ implies $a \leq c$;
- antisymmetric: $a \leq b$ and $b \leq a$ imply $a = b$;
- total: either $a \leq b$ or $b \leq a$.

Therefore $(\mathbb{N}, \leq)$ is a linearly ordered set.
The gap definition also gives, without further work,

$$a \leq a + b \qquad \text{and} \qquad a \leq b + a,$$

because addition is commutative.

# Definition of strict inequality

## Definition

For $a, b \in \mathbb{N}$,

$$a < b \iff a \leq b \text{ and } b \nleq a.$$

## Equivalent form

$$a < b \iff a \leq b \text{ and } a \neq b.$$

Indeed, if $a < b$ and $a = b$, reflexivity would give $b \leq a$, a contradiction. Conversely, if $a \leq b$, $a \neq b$, and $b \leq a$, antisymmetry would force $a = b$, again a contradiction.

# Positive numbers

## Proposition

For every $a \in \mathbb{N}$,

$$a \neq 0 \iff 0 < a.$$

## Proof.

We always have $0 \leq a$. If $a \neq 0$, the equivalent form of strict inequality therefore gives $0 < a$. Conversely, if $0 < a$, then strict inequality implies $0 \neq a$, which is the same as $a \neq 0$. $\square$

Thus, for natural numbers, nonzero and positive mean exactly the same thing.

# Strict inequality means a positive gap

**Theorem**

For $a, b \in \mathbb{N}$,
$$a < b \iff \text{there is an } x \neq 0 \text{ such that } b = a + x.$$

If $a < b$, choose $x$ with $b = a + x$. If $x = 0$, then $b = a + 0 = a$, contradicting $a \neq b$. Hence $x \neq 0$.
Conversely, suppose $b = a + x$ with $x \neq 0$. Then $a \leq b$. If $a = b$, we would have

$$a + 0 = a = b = a + x.$$

Cancellation would give $x = 0$, a contradiction. Thus $a \neq b$, and consequently $a < b$.

# Addition preserves order

## Theorem

If $a \leq b$, then $a + c \leq b + c$.

Choose $x \in \mathbb{N}$ with $b = a + x$. We look for a gap from $a + c$ to $b + c$. Using associativity and commutativity,

$$\begin{aligned} b + c &= (a + x) + c \\ &= a + (x + c) \\ &= a + (c + x) \\ &= (a + c) + x. \end{aligned}$$

Thus the same $x$ is a gap from $a + c$ to $b + c$, proving $a + c \leq b + c$.
By commutativity, adding $c$ on the left also preserves order.

# Cancelling the same summand from an inequality

**Theorem**

If $a + b \leq a + c$, then $b \leq c$.

Choose $x \in \mathbb{N}$ such that

$$a + c = (a + b) + x.$$

By associativity, the right-hand side is $a + (b + x)$, so

$$a + c = a + (b + x).$$

Addition cancellation gives

$$c = b + x.$$

Hence $x$ is a gap from $b$ to $c$, and $b \leq c$.
Together with order preservation, this yields

$$a + b \leq a + c \iff b \leq c.$$

# Positive multiplication preserves strict order

## Theorem

If $0 < c$ and $a < b$, then $ca < cb$.

Because $a < b$, write $b = a + x$ with $x \neq 0$. Because $0 < c$, we also have $c \neq 0$. Distributivity gives

$$cb = c(a + x) = ca + cx.$$

Since $c \neq 0$ and $x \neq 0$, the no-zero-divisors theorem gives $cx \neq 0$. Thus $cb$ is obtained from $ca$ by adding the nonzero gap $cx$. The positive-gap characterization now gives

$$ca < cb.$$

# Multiplication on the right

## Corollary

If $0 < c$ and $a < b$, then $ac < bc$.

By commutativity of multiplication,

$$ac = ca < cb = bc.$$

## Why positivity is necessary

If $c = 0$, then $ac = 0 = bc$, even when $a < b$. Multiplication by zero destroys strict inequality.

For weak inequalities, no positivity assumption is needed: if $a \leq b$, write $b = a + x$; then $bc = ac + xc$, so $ac \leq bc$.

# Cancellation for multiplication: reduction to order

**Theorem**

Suppose $a \neq 0$. If $ab = ac$, then $b = c$.

Since the order is total, either $b \leq c$ or $c \leq b$.
Assume first that $b \leq c$. If $b \neq c$, then $b < c$. Since $a \neq 0$ is equivalent to $0 < a$, strict order preservation gives

$$ab < ac,$$

contradicting $ab = ac$. Therefore $b = c$.
The case $c \leq b$ is symmetric: if $c \neq b$, then $c < b$, hence $ac < ab$, again contradicting the assumed equality.

# Right cancellation for multiplication

## Corollary

Suppose $a \neq 0$. If $ba = ca$, then $b = c$.

Commutativity converts the hypothesis into

$$ab = ac.$$

The multiplication cancellation theorem therefore gives $b = c$.

## Why $a \neq 0$ is necessary

For $a = 0$, the equality $b0 = c0$ holds for every $b, c \in \mathbb{N}$, so no conclusion about $b$ and $c$ is possible.

# Final picture

Starting from the natural-number principles, we proved that

- addition and multiplication obey the usual associative, commutative, distributive, and identity laws;
- sums and products have strong cancellation properties;
- nonzero products are nonzero;
- the relation $a \leq b \iff b = a + x$ is a linear order;
- $a < b$ means exactly that the gap from $a$ to $b$ is positive;
- addition preserves order, and multiplication by a positive number preserves strict order.

# An equation without solutions in $\mathbb{N}$

Suppose that some $x \in \mathbb{N}$ satisfied

$$3 + x = 1.$$

By commutativity and the recursive definition of addition,

$$3 + x = x + 3 = S(S(S(x))), \qquad 1 = S(0).$$

Hence

$$S(S(S(x))) = S(0).$$

Injectivity of the successor operation would then give

$$S(S(x)) = 0,$$

which is impossible because zero is not a successor. Therefore $3 + x = 1$ has no solution in $\mathbb{N}$.

# The integers

# Why introduce the integers?

In $\mathbb{N}$, the equation

$$3 + x = 1$$

has no solution. We want a larger number system in which it has a solution (namely $x = -2$). Every integer can be thought of as a difference

$$a - b, \qquad a, b \in \mathbb{N}.$$

Since subtraction is not yet available, we begin with the ordered pair $(a, b)$, intended to represent that difference.

## The central difficulty

The same integer has many representations:

$$5 - 2 = 4 - 1 = 3 - 0.$$

We must decide exactly when two pairs represent the same difference.

# The pre-integers

## Definition

The set of pre-integers is

$$\mathbb{Z}_{\mathrm{pre}} = \mathbb{N}^2,$$

the set of ordered pairs $(a, b)$ of natural numbers.

The pair $(a, b)$ records the difference we intend to create:

$$(a, b) \quad \longleftrightarrow \quad a - b.$$

Every pair is allowed: a difference makes sense for any $a, b \in \mathbb{N}$. The only problem, addressed next, is that different pre-integers can represent the same difference.

## Equality of formal differences

### Definition

For pre-integers $(a, b), (c, d) \in \mathbb{Z}_{\mathrm{pre}}$, declare

$$(a, b) \sim (c, d) \quad \iff \quad a + d = b + c.$$

The equation uses only addition in $\mathbb{N}$, but it expresses

$$a - b = c - d.$$

Indeed, cross-adding $b + d$ to the informal equation gives precisely $a + d = b + c$.
For example,

$$(5, 2) \sim (4, 1) \qquad \text{because} \qquad 5 + 1 = 2 + 4.$$

# Reflexivity and symmetry

## Proposition

The relation $\sim$ is reflexive and symmetric.

## Proof.

For reflexivity, let $(a, b) \in \mathbb{Z}_{\mathrm{pre}}$. The equality $a + b = b + a$ shows directly that $(a, b) \sim (a, b)$. For symmetry, suppose $(a, b) \sim (c, d)$, so $a + d = b + c$. Then

$$c + b = b + c = a + d = d + a.$$

The first and last equalities use commutativity. Thus $c + b = d + a$, which is exactly $(c, d) \sim (a, b)$. $\square$

# Transitivity

## Proposition

If $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then $(a, b) \sim (e, f)$.

## Proof.

The assumptions say

$$a + d = b + c, \qquad c + f = d + e.$$

Add the two equalities and rearrange:

$$(a + f) + (c + d) = (b + e) + (c + d).$$

Addition in $\mathbb{N}$ admits cancellation, so cancelling the common summand $c + d$ yields

$$a + f = b + e.$$

This is precisely the condition $(a, b) \sim (e, f)$. $\square$

# The integers

Reflexivity, symmetry, and transitivity show that $\sim$ is an equivalence relation on the pre-integers.

## Definition

The set of integers is the set of equivalence classes

$$\boxed{\mathbb{Z} = \mathbb{Z}_{\text{pre}}/\sim}.$$

We write

$$[(a, b)]$$

for the class represented by $(a, b)$.

Thus the equality criterion in $\mathbb{Z}$ is

$$\boxed{[(a, b)] = [(c, d)] \iff a + d = b + c.}$$

This criterion will be used repeatedly.

# The universal property of a quotient

Let $X$ be a set with an equivalence relation $\sim$, let $Y$ be a set, and write $q\colon X \to X/{\sim}$, $q(x) = [x]$, for the quotient map.

## Universal property

A function $f\colon X \to Y$ descends to a function $\bar{f}\colon X/{\sim} \to Y$ satisfying

$$\bar{f}([x]) = f(x)$$

if and only if it is constant on equivalence classes:

$$x \sim x' \quad \Longrightarrow \quad f(x) = f(x').$$

When it exists, $\bar{f}$ is unique; equivalently, $f = \bar{f} \circ q$.

Thus functions $X/{\sim} \to Y$ correspond exactly to functions $X \to Y$ that are constant on equivalence classes.

## How to define a function on a quotient

Give a formula on representatives and prove that equivalent representatives have the same image. The descended function is then uniquely determined.

# Negation and addition

The familiar rules for differences suggest the definitions

$$-(a-b) = b-a, \qquad (a-b)+(c-d) = (a+c)-(b+d).$$

## Definitions

For integer classes, set

$$-[(a,b)] = [(b,a)], \qquad [(a,b)] + [(c,d)] = [(a+c, b+d)].$$

Before using these formulas, we must prove that their values do not depend on the chosen representatives.

# Negation is well defined

**Proposition**

If $(a, b) \sim (c, d)$, then $(b, a) \sim (d, c)$.

**Proof.**

The assumption is

$$a + d = b + c.$$

To prove $(b, a) \sim (d, c)$, we need

$$b + c = a + d,$$

which is the same equality read in the opposite direction. $\square$

Hence equivalent pairs have equivalent negatives, so $-[(a, b)]$ is independent of the representative $(a, b)$.

# Addition is well defined

Suppose

$$(a,b) \sim (a',b'), \qquad (c,d) \sim (c',d').$$

Thus $a + b' = b + a'$ and $c + d' = d + c'$. Adding these equations and rearranging gives

$$(a+c) + (b'+d') = (b+d) + (a'+c').$$

Therefore

$$(a+c, b+d) \sim (a'+c', b'+d').$$

We have proved that replacing either summand by an equivalent pair does not alter the class of the sum. Consequently,

$$[(a,b)] + [(c,d)] = [(a+c, b+d)]$$

is a well-defined operation on $\mathbb{Z}$.

# Discovering multiplication

Formally expanding a product of differences gives

$$(a - b)(c - d) = (ac + bd) - (ad + bc).$$

This suggests the following definition.

**Definition**

$$\boxed{[(a, b)]\,[(c, d)] = [(ac + bd, ad + bc)]\,.}$$

For example,

$$[(5, 2)]\,[(1, 4)] = [(5 \cdot 1 + 2 \cdot 4, 5 \cdot 4 + 2 \cdot 1)] = [(13, 22)]\,,$$

which represents $-9$, as expected from $3 \cdot (-3) = -9$.

# Multiplication respects one representative

Suppose $(a, b) \sim (c, d)$, so $a + d = b + c$, and fix $(e, f)$. Multiplying the given equality by $e$ and by $f$ gives

$$ae + de = be + ce, \qquad af + df = bf + cf.$$

We must compare

$$(ae + bf, af + be) \quad \text{and} \quad (ce + df, cf + de).$$

Their cross-sums satisfy

$$\begin{aligned}(ae + bf) + (cf + de) &= (ae + de) + (bf + cf)\\ &= (be + ce) + (af + df)\\ &= (af + be) + (ce + df).\end{aligned}$$

Hence the two product pairs are equivalent.

# Multiplication is well defined

Pair multiplication is commutative:

$$(ae + bf\,, af + be) = (ea + fb, eb + fa),$$

by commutativity in $\mathbb{N}$.
The preceding argument therefore works in either factor. If

$$(a, b) \sim (a', b'), \qquad (c, d) \sim (c', d'),$$

then

$$(a, b)(c, d) \sim (a', b')(c, d) \sim (a', b')(c', d').$$

Transitivity now shows that the product class is unchanged when both representatives are replaced.
Thus multiplication is a well-defined operation on $\mathbb{Z}$.

# Zero and one

Define

$$0 = [(0,0)]\,, \qquad 1 = [(1,0)]\,.$$

The equality criterion immediately gives a useful test:

$$\boxed{[(a,b)] = 0 \quad \Longleftrightarrow \quad a = b.}$$

Indeed,

$$[(a,b)] = [(0,0)] \quad \Longleftrightarrow \quad a+0 = b+0 \quad \Longleftrightarrow \quad a = b.$$

In particular $0 \neq 1$, since $0 \neq 1$ in $\mathbb{N}$.

# Associativity of addition

## Theorem

For all $x, y, z \in \mathbb{Z}$, $(x + y) + z = x + (y + z)$.

## Proof.

Choose representatives

$$x = [(a, b)]\,, \qquad y = [(c, d)]\,, \qquad z = [(e, f)]\,.$$

Then

$$\begin{aligned} (x + y) + z &= [((a + c) + e, (b + d) + f)]\,, \\ x + (y + z) &= [(a + (c + e), b + (d + f))]\,. \end{aligned}$$

The two displayed pairs are equal by associativity of addition in $\mathbb{N}$. Therefore their integer classes are equal. $\square$

# The additive identity and additive inverses

Let $x = [(a, b)]$.

## Zero is an identity

$$x + 0 = [(a + 0, b + 0)] = [(a, b)] = x.$$

Commutativity of natural addition similarly gives $0 + x = x$.

## Every integer has an additive inverse

$$x + (-x) = [(a + b, b + a)].$$

Since $a + b = b + a$, the zero criterion gives

$$x + (-x) = 0.$$

# Commutativity and subtraction

If $x = [(a,b)]$ and $y = [(c,d)]$, then

$$x + y = [(a+c, b+d)] = [(c+a, d+b)] = y + x.$$

Thus addition on $\mathbb{Z}$ is commutative.

## Definition of subtraction

We may now define

$$x - y = x + (-y).$$

In terms of representatives,

$$[(a,b)] - [(c,d)] = [(a+d, b+c)] .$$

The operation that motivated the construction has now become available.

# The multiplicative identity and commutativity

Let $x = [(a,b)]$. Then

$$1x = [(1,0)]\,[(a,b)] = [(1a + 0b, 1b + 0a)] = [(a,b)] = x.$$

If $y = [(c,d)]$, then

$$\begin{aligned} xy &= [(ac + bd, ad + bc)]\,, \\ yx &= [(ca + db, cb + da)]\,. \end{aligned}$$

Corresponding coordinates are equal by commutativity in $\mathbb{N}$, so $xy = yx$. Consequently $x1 = 1x = x$ as well.

# Associativity of multiplication: first coordinates

Let

$$x = [(a,b)]\,, \qquad y = [(c,d)]\,, \qquad z = [(e,f)]\,.$$

The first coordinate of a representative of $(xy)z$ is

$$(ac + bd)e + (ad + bc)f = ace + bde + adf + bcf\,.$$

The first coordinate of a representative of $x(yz)$ is

$$a(ce + df) + b(cf + de) = ace + adf + bcf + bde.$$

These are equal by associativity, commutativity, and distributivity in $\mathbb{N}$.

# Associativity of multiplication: second coordinates

With the same notation, the second coordinate of $(xy)z$ is

$$(ac + bd)f + (ad + bc)e = acf + bdf + ade + bce.$$

The second coordinate of $x(yz)$ is

$$a(cf + de) + b(ce + df) = acf + ade + bce + bdf.$$

Again the two expressions are equal after rearranging their summands. Therefore the representing pairs are equal, and

$$\boxed{(xy)z = x(yz).}$$

# Distributivity

Let $x = [(a,b)]$, $y = [(c,d)]$, and $z = [(e,f)]$. The first coordinate of $x(y+z)$ is

$$a(c+e) + b(d+f) = ac + ae + bd + bf,$$

while that of $xy + xz$ is

$$(ac + bd) + (ae + bf).$$

They are equal after rearrangement. The second coordinates are likewise

$$a(d+f) + b(c+e) = ad + af + bc + be$$

and

$$(ad + bc) + (af + be).$$

Hence $x(y+z) = xy + xz$. Commutativity of multiplication gives $(x+y)z = xz + yz$.

# The algebraic structure of $\mathbb{Z}$

We have proved that

- addition is associative and commutative, with identity 0;
- every integer has an additive inverse;
- multiplication is associative and commutative, with identity 1;
- multiplication distributes over addition;
- $0 \neq 1$.

Therefore

$$(\mathbb{Z}, 0, 1, +, -, \cdot)$$

is a commutative ring.

Every one of these laws ultimately follows from the arithmetic of $\mathbb{N}$ and the equality criterion for integer classes.

# The rule of signs

The classes $[(n, 0)]$ and $[(0, n)] = -[(n, 0)]$ represent a natural number and its negative. Applying the multiplication formula directly gives, for $m, n \in \mathbb{N}$,

| $\cdot$ | $[(n, 0)]$ | $[(0, n)]$ |
|---|---|---|
| $[(m, 0)]$ | $[(mn, 0)]$ | $[(0, mn)]$ |
| $[(0, m)]$ | $[(0, mn)]$ | $[(mn, 0)]$ |

This is the familiar rule of signs:

$$(+)(+) = +, \quad (+)(-) = -, \quad (-)(+) = -, \quad (-)(-) = +.$$

The table illustrates the signs of these particular representatives.

# A natural-number lemma

## Lemma

If $a, b, c, d \in \mathbb{N}$ satisfy

$$ad + bc = ac + bd,$$

then $a = b$ or $c = d$.

The proof follows the induction used in the natural-number development. Induct on $a$, allowing $b$ to vary. If $a = 0$, the equality is $bc = bd$, so either $b = 0 = a$ or cancellation gives $c = d$. If $a$ is a successor, split into the cases $b = 0$ and $b = S(f)$. The first again gives $c = d$. In the second, expanding both sides and cancelling the common terms $c + d$ reduces the equality to the induction hypothesis.

# No zero divisors

## Theorem

If $x \neq 0$ and $y \neq 0$, then $xy \neq 0$.

## Proof.

Choose arbitrary representatives

$$x = [(a, b)], \qquad y = [(c, d)].$$

If $xy = 0$, the multiplication formula and the equality criterion give

$$ad + bc = ac + bd.$$

The natural-number lemma yields $a = b$ or $c = d$. In the first case $x = [(a, a)] = 0$; in the second, $y = [(c, c)] = 0$. Both conclusions contradict the hypotheses, so $xy \neq 0$. $\square$

# Cancellation for multiplication

**Theorem**

If $x \neq 0$ and $yx = zx$, then $y = z$.

**Proof.**

Subtract $zx$ from both sides:

$$yx - zx = 0.$$

By distributivity,

$$(y - z)x = 0.$$

If $y - z \neq 0$, then both factors would be nonzero, contradicting the no-zero-divisors theorem. Hence $y - z = 0$, which is equivalent to $y = z$. $\square$

The assumption $x \neq 0$ is necessary, since $y0 = z0$ for all $y, z$.

# The natural embedding

**Definition**

Define

$$\iota\colon \mathbb{N} \longrightarrow \mathbb{Z}, \qquad \iota(n) = [(n, 0)]\,.$$

This map sends a natural number to the integer represented by $n - 0$. It preserves the distinguished elements:

$$\iota(0) = [(0,0)] = 0, \qquad \iota(1) = [(1,0)] = 1.$$

# The embedding preserves arithmetic

For $a, b \in \mathbb{N}$,

$$\begin{aligned} \iota(a) + \iota(b) &= [(a, 0)] + [(b, 0)] \\ &= [(a + b, 0)] = \iota(a + b), \end{aligned}$$

and

$$\begin{aligned} \iota(a)\iota(b) &= [(a, 0)]\,[(b, 0)] \\ &= [(ab + 0 \cdot 0, a \cdot 0 + 0 \cdot b)] \\ &= [(ab, 0)] = \iota(ab). \end{aligned}$$

Thus natural addition and multiplication agree with the integer operations.

# The embedding is injective

## Proposition

If $\iota(a) = \iota(b)$, then $a = b$.

## Proof.

The equality

$$[(a, 0)] = [(b, 0)]$$

is equivalent to

$$a + 0 = 0 + b.$$

Hence $a = b$. $\square$

We may therefore identify $\mathbb{N}$ with its image in $\mathbb{Z}$, obtaining the familiar inclusion

$$\mathbb{N} \subseteq \mathbb{Z}.$$

# Order as a nonnegative gap

**Definition**

For $x, y \in \mathbb{Z}$, define

$$\boxed{x \leq y \quad \Longleftrightarrow \quad y = x + \iota(n) \text{ for some } n \in \mathbb{N}.}$$

Thus $x \leq y$ means that the gap $y - x$ is a natural number.
Immediate examples are

$$x \leq x \quad \text{using } n = 0, \qquad 0 \leq 1 \quad \text{using } n = 1.$$

# Reflexivity of the integer order

## Proposition

For every $x \in \mathbb{Z}$, one has $x \leq x$.

Choose $0 \in \mathbb{N}$ as the witness in the definition of $\leq$. Since $\iota(0) = 0$,

$$x = x + 0 = x + \iota(0).$$

Therefore $x \leq x$.

# Transitivity of the integer order

Suppose $x \leq y$ and $y \leq z$. Unpacking the two hypotheses gives $p, q \in \mathbb{N}$ such that

$$y = x + \iota(p), \qquad z = y + \iota(q).$$

Substitute these equalities and use $p + q$ as the new witness:

$$\begin{aligned} z &= (x + \iota(p)) + \iota(q) \\ &= x + \big(\iota(p) + \iota(q)\big) \\ &= x + \iota(p + q). \end{aligned}$$

Hence $x \leq z$.

# Antisymmetry: the two gaps

Suppose $x \leq y$ and $y \leq x$. Choose $p, q \in \mathbb{N}$ such that

$$y = x + \iota(p), \qquad x = y + \iota(q).$$

Substitute $y = x + \iota(p)$ in the second equality. Then

$$x = (x + \iota(p)) + \iota(q) = x + \iota(p + q).$$

Cancelling $x$ from the additive equality yields

$$\iota(p + q) = 0 = \iota(0).$$

# Antisymmetry: the gaps vanish

Since $\iota$ is injective,

$$\iota(p+q) = \iota(0) \quad \Longrightarrow \quad p+q=0.$$

A sum of natural numbers is zero only when both summands are zero, so

$$p = 0, \qquad q = 0.$$

Therefore

$$y = x + \iota(p) = x + \iota(0) = x.$$

We have proved that $x \leq y$ and $y \leq x$ imply $x = y$.

# Totality: the first natural comparison

Let

$$x = [(a, b)]\,, \qquad y = [(c, d)]\,.$$

The natural numbers $a + d$ and $b + c$ are comparable:

$$a + d \leq b + c \qquad \text{or} \qquad b + c \leq a + d.$$

Suppose first that $a + d \leq b + c$. By the definition of the order on $\mathbb{N}$, choose $e \in \mathbb{N}$ such that

$$b + c = (a + d) + e$$

and use $e$ as a witness for $x \leq y$. Indeed,

$$x + \iota(e) = [(a + e, b)]\,.$$

The displayed natural-number equality, after reassociation and commutation, is

$$c + b = d + (a + e).$$

Thus $[(c, d)] = [(a + e, b)] = x + \iota(e)$, so $x \leq y$.

# Totality: the second natural comparison

Now suppose that $b + c \leq a + d$. Choose $e \in \mathbb{N}$ such that

$$a + d = (b + c) + e.$$

This time use $e$ as a witness for $y \leq x$. We have

$$y + \iota(e) = [(c + e, d)],$$

while reassociating the chosen equality gives

$$a + d = b + (c + e).$$

Hence

$$[(a, b)] = [(c + e, d)] = y + \iota(e),$$

and therefore $y \leq x$. In either case, $x \leq y$ or $y \leq x$.

# The integers form a linear order

We have proved, directly from the witness definition of $\leq$, that for all $x, y, z \in \mathbb{Z}$:

$$\begin{aligned}
&x \leq x, \\
&x \leq y \text{ and } y \leq z \quad \Longrightarrow \quad x \leq z, \\
&x \leq y \text{ and } y \leq x \quad \Longrightarrow \quad x = y, \\
&x \leq y \text{ or } y \leq x.
\end{aligned}$$

## Conclusion

These are exactly the defining properties of a linear order. Thus $(\mathbb{Z}, \leq)$ is a linear order.

# The embedding preserves and reflects order

## Proposition

For $a, b \in \mathbb{N}$,

$$\boxed{\iota(a) \leq \iota(b) \iff a \leq b.}$$

Suppose $\iota(a) \leq \iota(b)$. Choose $n \in \mathbb{N}$ such that

$$\iota(b) = \iota(a) + \iota(n) = \iota(a + n).$$

Injectivity of $\iota$ gives $b = a + n$, hence $a \leq b$.
Conversely, if $a \leq b$, choose $k \in \mathbb{N}$ such that $b = a + k$. Then

$$\iota(b) = \iota(a + k) = \iota(a) + \iota(k),$$

so $k$ witnesses $\iota(a) \leq \iota(b)$.

# Addition preserves order

**Theorem**

If $x \leq y$, then $x + z \leq y + z$.

Choose $n \in \mathbb{N}$ such that $y = x + \iota(n)$. Then

$$\begin{aligned} y + z &= (x + \iota(n)) + z \\ &= (x + z) + \iota(n), \end{aligned}$$

where associativity and commutativity of addition justify the rearrangement. Thus the same natural gap $n$ proves

$$x + z \leq y + z.$$

# Strict positivity

Since the order is linear, we write

$$x < y \iff x \leq y \text{ and } x \neq y.$$

If $0 < x$, then $0 \leq x$, so the definition of order gives

$$x = 0 + \iota(n) = \iota(n)$$

for some $n \in \mathbb{N}$. Moreover $n \neq 0$, because $x \neq 0$ and $\iota$ is injective.
Conversely, if $n \neq 0$, then

$$0 < \iota(n).$$

Indeed $0 \leq \iota(n)$ has gap $n$, and injectivity gives $\iota(n) \neq 0$.
Therefore the positive integers are exactly the images of the positive natural numbers.

# A product of positive integers is positive

**Theorem**

If $0 < x$ and $0 < y$, then $0 < xy$.

Write

$$x = \iota(n), \qquad y = \iota(m), \qquad n, m \neq 0.$$

Since the embedding preserves multiplication,

$$xy = \iota(n)\iota(m) = \iota(nm).$$

The natural-number product $nm$ is nonzero. The characterization of positive integers therefore gives

$$0 < \iota(nm) = xy.$$

# An ordered commutative ring

We have shown that

- $\mathbb{Z}$ is a commutative ring;
- $\leq$ is a linear order;
- adding the same integer preserves order;
- the product of two positive integers is positive.

These facts make $\mathbb{Z}$ an ordered commutative ring.
They also recover all the familiar sign behavior. For example, translating $x < y$ by $-x$ gives

$$0 < y - x.$$

# The Archimedean bound

## Theorem

For every $x \in \mathbb{Z}$, there is an $n \in \mathbb{N}$ such that

$$x \leq \iota(n).$$

Let $x = [(a, b)]$. We claim that $n = a$ works. Indeed,

$$x + \iota(b) = [(a, b)] + [(b, 0)] = [(a + b, b)] \,.$$

But

$$[(a + b, b)] = [(a, 0)] = \iota(a),$$

because $(a + b) + 0 = b + a$. Thus

$$\iota(a) = x + \iota(b),$$

and the natural gap $b$ proves $x \leq \iota(a)$.

# Final picture

Starting from $\mathbb{N}$, we constructed $\mathbb{Z}$ and proved that

- integers are equivalence classes of pre-integers, that is, of formal differences;
- addition, negation, and multiplication are independent of representatives;
- $\mathbb{Z}$ is a commutative ring with no zero divisors;
- nonzero factors may be cancelled;
- $\mathbb{N}$ embeds into $\mathbb{Z}$, preserving arithmetic and order;
- the nonnegative-gap relation is a linear order;
- addition preserves order and positive products are positive;
- every integer is bounded above by a natural number.

# An equation without solutions in $\mathbb{Z}$

Suppose that some $x \in \mathbb{Z}$ satisfied

$$2x = 1.$$

By totality of the integer order,

$$0 \leq x \qquad \text{or} \qquad x \leq 0.$$

If $0 \leq x$, write $x = \iota(n)$ with $n \in \mathbb{N}$. Since $\iota$ preserves multiplication and is injective,

$$\iota(2n) = 2\iota(n) = 2x = 1 = \iota(1).$$

Hence

$$2n = 1.$$

If $x \leq 0$, write $0 = x + \iota(n)$ with $n \in \mathbb{N}$. Multiplying by 2 and again using injectivity gives

$$0 = 2x + \iota(2n) = 1 + \iota(2n) = \iota(1 + 2n),$$

and hence

$$0 = 1 + 2n.$$

# An equation without solutions in $\mathbb{Z}$ (continued)

We now rule out the two natural-number equations exactly as before, using the recursive description of addition and the successor principles.
First,

$$1 + 2n = 2n + 1 = S((2n)),$$

so $0 = 1 + 2n$ is impossible because zero is not a successor.
Now suppose that $2n = 1$. If $n = 0$, this says $0 = S(0)$, which is impossible. If $n = S(m)$, then

$$2n = 2(m + 1) = 2m + 2 = S(S((2m))).$$

Hence $2n = 1 = S(0)$ would imply, by injectivity of the successor operation,

$$S((2m)) = 0,$$

again impossible because zero is not a successor.
Therefore neither natural-number equation can hold, and $2x = 1$ has no solution in $\mathbb{Z}$.

# The rational numbers

# Why introduce the rational numbers?

In $\mathbb{Z}$, the equation

$$2x = 1$$

has no solution. To solve it we want to introduce fractions $a/b$, where $a, b \in \mathbb{Z}$ and $b \neq 0$. We begin with a pair $(a, b)$, intended to represent $a/b$.

## The central difficulty

A rational number has many representatives:

$$\frac{1}{2} = \frac{2}{4} = \frac{-3}{-6}.$$

Before doing arithmetic, we must say exactly when two pairs represent the same number.

# The pre-rationals

## Definition

The set of pre-rationals is

$$\mathbb{Q}_{\mathrm{pre}} = \{(a, b) \in \mathbb{Z}^2 : b \neq 0\}.$$

The restriction on the denominator is essential. It ensures that products of denominators remain nonzero and, later, that cross multiplication admits cancellation.
We shall use fraction notation for the pair when no confusion can arise:

$$(a, b) \quad \longleftrightarrow \quad \frac{a}{b}.$$

# When do two fractions agree?

## Definition

For $(a, b), (c, d) \in \mathbb{Q}_{\mathrm{pre}}$, declare

$$(a, b) \sim (c, d) \quad \Longleftrightarrow \quad ad = bc.$$

This uses only multiplication and equality in $\mathbb{Z}$, but it captures the familiar rule

$$\frac{a}{b} = \frac{c}{d} \quad \Longleftrightarrow \quad ad = bc.$$

For example, $(1, 2) \sim (-3, -6)$ because $1 \cdot (-6) = 2 \cdot (-3)$.

# Reflexivity and symmetry

## Proposition

The relation $\sim$ is reflexive and symmetric.

## Proof.

For every $(a, b)$, commutativity gives $ab = ba$, hence $(a, b) \sim (a, b)$.
If $(a, b) \sim (c, d)$, then $ad = bc$. Reading the equality backwards and commuting both products gives

$$cb = da,$$

which says $(c, d) \sim (a, b)$. $\square$

# Transitivity uses the nonzero denominator

Suppose

$$(a, b) \sim (c, d), \qquad (c, d) \sim (e, f).$$

Thus $ad = bc$ and $cf = de$. Multiplying the first equality by $f$ and the second by $b$ gives

$$adf = bcf = bde.$$

After rearranging,

$$d(af) = d(be).$$

Since $d \neq 0$, cancellation in $\mathbb{Z}$ yields $af = be$. Therefore $(a, b) \sim (e, f)$.

**Why $b \neq 0$ belongs in the definition**

Without a nonzero middle denominator, this cancellation argument would fail and transitivity would not follow.

# The rational numbers

We have proved that $\sim$ is an equivalence relation.

## Definition

The rational numbers are the equivalence classes

$$\boxed{\mathbb{Q} = \mathbb{Q}_{\mathrm{pre}}/\!\sim}.$$

We write

$$\left[\frac{a}{b}\right]$$

for the class represented by $(a, b)$.

Equality in $\mathbb{Q}$ is therefore governed by the single criterion

$$\boxed{\left[\frac{a}{b}\right] = \left[\frac{c}{d}\right] \quad \Longleftrightarrow \quad ad = bc.}$$

# The four basic operations

The usual fraction formulas suggest

$$-\left[\frac{a}{b}\right]=\left[\frac{-a}{b}\right],$$
$$\left[\frac{a}{b}\right]+\left[\frac{c}{d}\right]=\left[\frac{ad+bc}{bd}\right],$$
$$\left[\frac{a}{b}\right]\left[\frac{c}{d}\right]=\left[\frac{ac}{bd}\right].$$

If $\left[\frac{a}{b}\right]\neq 0$, equivalently $a\neq 0$, we define its reciprocal by

$$\left(\left[\frac{a}{b}\right]\right)^{-1}=\left[\frac{b}{a}\right].$$

All displayed denominators are nonzero: this follows from the absence of zero divisors in $\mathbb{Z}$, and from $a\neq 0$ in the reciprocal formula. The next task is to prove that the answers do not depend on the chosen representatives.

# Addition is well defined

Suppose

$$\frac{a}{b} \sim \frac{a'}{b'}, \qquad \frac{c}{d} \sim \frac{c'}{d'},$$

so $ab' = ba'$ and $cd' = dc'$. We compare the proposed sums by cross multiplication:

$$\begin{aligned}(ad+bc)b'd' &= ab'dd' + bb'cd' \\ &= ba'dd' + bb'dc' \\ &= bd(a'd' + b'c').\end{aligned}$$

Hence

$$\frac{ad+bc}{bd} \sim \frac{a'd'+b'c'}{b'd'}.$$

Thus addition passes from pre-rationals to equivalence classes.

# Multiplication is well defined

Under the same hypotheses $ab' = ba'$ and $cd' = dc'$,

$$\begin{aligned}(ac)(b'd') &= (ab')(cd')\\ &= (ba')(dc')\\ &= (bd)(a'c').\end{aligned}$$

Consequently,

$$\frac{ac}{bd} \sim \frac{a'c'}{b'd'}.$$

Negation is even more direct: from $ad = bc$ we obtain

$$(-a)d = b(-c),$$

so $(-a, b) \sim (-c, d)$.
Therefore multiplication and negation are well defined on $\mathbb{Q}$.

# The zero class detects a zero numerator

**Proposition**

If $b \neq 0$, then

$$\left[\frac{a}{b}\right] = \left[\frac{0}{1}\right] \iff a = 0.$$

**Proof.**

The equality criterion says

$$a \cdot 1 = b \cdot 0,$$

which is equivalent to $a = 0$. $\square$

In particular, a nonzero rational has a nonzero numerator in every representative. This is exactly what is needed to define its reciprocal.

# The reciprocal is well defined

Suppose $(a, b) \sim (c, d)$ and that their common class is nonzero. Then $ad = bc$, and the preceding proposition gives

$$a \neq 0 \qquad \text{and} \qquad c \neq 0.$$

Thus $(b, a)$ and $(d, c)$ are valid pre-rationals. Moreover,

$$(b, a) \sim (d, c) \quad \Longleftrightarrow \quad bc = ad,$$

which is our original equality in reverse.
Therefore $x^{-1} = \left[\frac{b}{a}\right]$ is independent of the representative chosen for every nonzero $x \in \mathbb{Q}$.

# Zero, one, and subtraction

We define

$$0 = \left[\frac{0}{1}\right], \qquad 1 = \left[\frac{1}{1}\right].$$

Subtraction is addition of the negative. Directly from the fraction formulas,

$$\boxed{\left[\frac{a}{b}\right] - \left[\frac{c}{d}\right] = \left[\frac{ad - bc}{bd}\right].}$$

Also $0 \neq 1$: otherwise the equality criterion would give

$$0 \cdot 1 = 1 \cdot 1,$$

contradicting $0 \neq 1$ in $\mathbb{Z}$.

# The commutative-ring laws

Every ring identity can now be checked on representatives. For example,

$$\left(\left[\frac{a}{b}\right] + \left[\frac{c}{d}\right]\right) + \left[\frac{e}{f}\right] = \left[\frac{(ad+bc)f + (bd)e}{bdf}\right]$$
$$= \left[\frac{adf + bcf + bde}{bdf}\right],$$

$$\left[\frac{a}{b}\right] + \left(\left[\frac{c}{d}\right] + \left[\frac{e}{f}\right]\right) = \left[\frac{a(df) + b(cf+de)}{bdf}\right]$$
$$= \left[\frac{adf + bcf + bde}{bdf}\right].$$

Associativity and commutativity of both operations, the identity laws, additive inverses, and distributivity all reduce in this way to polynomial identities in $\mathbb{Z}$.

## Conclusion

The rational numbers form a nontrivial commutative ring.

# Multiplying by the reciprocal

## Proposition

If $x \in \mathbb{Q}$ and $x \neq 0$, then $xx^{-1} = 1$.

## Proof.

Write $x = \left[\frac{a}{b}\right]$. Since $x \neq 0$, its numerator satisfies $a \neq 0$. Therefore

$$xx^{-1} = \left[\frac{a}{b}\right]\left[\frac{b}{a}\right] = \left[\frac{ab}{ba}\right] = \left[\frac{1}{1}\right],$$

where the last equality follows from $(ab) \cdot 1 = (ba) \cdot 1$. $\square$

# The field of rational numbers

We have constructed:

- a nontrivial commutative ring $\mathbb{Q}$;
- a reciprocal $x^{-1}$ for every nonzero rational $x$;
- the identity $xx^{-1} = 1$ for every nonzero rational $x$.

**Theorem**

With these operations, $\mathbb{Q}$ is a field.

Thus for $y \neq 0$, division can be defined by

$$\frac{x}{y} = xy^{-1},$$

and equations such as $2x = 1$ now have the expected solution $x = 2^{-1}$.

# The integer embedding

## Definition

Define

$$j : \mathbb{Z} \longrightarrow \mathbb{Q}, \qquad j(a) = \left[\frac{a}{1}\right].$$

The fraction formulas show that

$$\begin{aligned} j(0) &= 0, & j(1) &= 1, \\ j(a+c) &= j(a) + j(c), & j(ac) &= j(a)j(c). \end{aligned}$$

Hence the old integer arithmetic is preserved inside the new number system.

# No integers are identified

**Proposition**

The map $j\colon \mathbb{Z} \to \mathbb{Q}$ is injective.

**Proof.**

If $j(a) = j(c)$, then

$$\left[\frac{a}{1}\right] = \left[\frac{c}{1}\right].$$

Cross multiplication gives $a \cdot 1 = 1 \cdot c$, hence $a = c$. $\square$

We may therefore regard $\mathbb{Z}$ as a subring of $\mathbb{Q}$, although the construction keeps the embedding visible.

# Every fraction is a quotient of integers

## Proposition

For $a, b \in \mathbb{Z}$ with $b \neq 0$,

$$\boxed{\left[\frac{a}{b}\right] = j(a)j(b)^{-1}.}$$

Indeed,

$$j(a)j(b)^{-1} = \left[\frac{a}{1}\right]\left[\frac{1}{b}\right] = \left[\frac{a}{b}\right].$$

This formula says that the quotient construction has done exactly what was intended: it adjoins inverses of all nonzero integers, and every rational is obtained from those inverses.

# The natural embedding is compatible

Let $\iota\colon \mathbb{N} \to \mathbb{Z}$ be the embedding already constructed for the integers. Define

$$i\colon \mathbb{N} \longrightarrow \mathbb{Q}, \qquad i(n) = \left[\frac{\iota(n)}{1}\right].$$

Then

$$i = j \circ \iota.$$

Since both earlier embeddings preserve $0, 1, +, \cdot$ and are injective, the same is true of $i$. Thus the chain

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q}$$

preserves the arithmetic constructed at every previous stage.

# Choosing a positive denominator

## Proposition

Every rational number has a representative $\left[\frac{a}{b}\right]$ with $b > 0$.

## Proof.

Begin with any representative $\left[\frac{a}{b}\right]$, where $b \neq 0$. The total order on $\mathbb{Z}$ gives either $b > 0$ or $b < 0$.

In the first case there is nothing to do. In the second case, $-b > 0$ and

$$\left[\frac{a}{b}\right] = \left[\frac{-a}{-b}\right],$$

because $a(-b) = b(-a)$. $\square$

Positive denominators let the sign of a rational be read from its numerator.

# Definition of nonnegativity

## Definition

A rational $x$ is nonnegative if there are integers $a, b$ such that

$$a \geq 0, \qquad b > 0, \qquad x = \left[\frac{a}{b}\right].$$

We write this property temporarily as $x \in \mathbb{Q}_{\geq 0}$. It is independent of any single chosen representative: it asks only for the existence of one representative whose numerator and denominator have the required signs.

Clearly $0, 1 \in \mathbb{Q}_{\geq 0}$, represented by $0/1$ and $1/1$.

# Every rational has one of the two signs

## Proposition

For every $x \in \mathbb{Q}$, either $x \in \mathbb{Q}_{\geq 0}$ or $-x \in \mathbb{Q}_{\geq 0}$.

## Proof.

Choose $x = \left[\frac{a}{b}\right]$ with $b > 0$. By totality in $\mathbb{Z}$, either $a \geq 0$ or $a < 0$.
In the first case, $a/b$ witnesses that $x$ is nonnegative. In the second, $-a > 0$, and

$$-x = \left[\frac{-a}{b}\right]$$

is nonnegative. $\square$

# Opposite signs meet only at zero

## Proposition

If both $x$ and $-x$ are nonnegative, then $x = 0$.

## Proof.

Choose

$$x = \left[\frac{a}{b}\right], \qquad -x = \left[\frac{c}{d}\right],$$

with $a, c \geq 0$ and $b, d > 0$. Since $-x = \left[\frac{-a}{b}\right]$, equality of the second pair of representatives gives

$$(-a)d = bc.$$

The left side is nonpositive and the right side is nonnegative, so both are zero. Since $d > 0$, this forces $a = 0$, and hence $x = 0$. $\square$

# Nonnegativity is closed under addition

Suppose

$$x = \left[\frac{a}{b}\right], \qquad y = \left[\frac{c}{d}\right],$$

where $a, c \geq 0$ and $b, d > 0$. Then

$$x + y = \left[\frac{ad + bc}{bd}\right].$$

Products of nonnegative integers are nonnegative, so $ad + bc \geq 0$. Products of positive integers are positive, so $bd > 0$.

## Conclusion

If $x, y \in \mathbb{Q}_{\geq 0}$, then $x + y \in \mathbb{Q}_{\geq 0}$.

# Nonnegativity and multiplication

With the same positive-denominator representatives,

$$xy = \left[\frac{ac}{bd}\right].$$

Since $ac \geq 0$ and $bd > 0$, the product is nonnegative.
If $x \neq 0$, then $a > 0$, and

$$x^{-1} = \left[\frac{b}{a}\right]$$

is positive because $b > 0$ and $a > 0$.

## Conclusion

Products of nonnegative rationals are nonnegative, and reciprocals of positive rationals are positive.

# Order as a nonnegative gap

The order on $\mathbb{Z}$ was defined by asking whether one integer differs from another by a nonnegative amount. We use the same idea here.

**Definition**

For $x, y \in \mathbb{Q}$, define

$$\boxed{x \leq y \quad \Longleftrightarrow \quad y - x \in \mathbb{Q}_{\geq 0}.}$$

In particular,

$$0 \leq x \quad \Longleftrightarrow \quad x \in \mathbb{Q}_{\geq 0},$$

because $x - 0 = x$.

# Comparing positive-denominator fractions

## Proposition

If $b, d > 0$, then

$$\boxed{\left[\frac{a}{b}\right] \leq \left[\frac{c}{d}\right] \quad \Longleftrightarrow \quad ad \leq bc.}$$

Indeed,

$$\left[\frac{c}{d}\right] - \left[\frac{a}{b}\right] = \left[\frac{bc - ad}{bd}\right],$$

and $bd > 0$. Thus the gap is nonnegative exactly when $bc - ad \geq 0$, which is equivalent to $ad \leq bc$. This recovers the ordinary cross-multiplication rule without ever reversing an inequality, because the denominators have first been made positive.

# Reflexivity and transitivity

## Proposition

The relation $\leq$ on $\mathbb{Q}$ is reflexive and transitive.

For reflexivity,

$$x - x = 0 \in \mathbb{Q}_{\geq 0},$$

so $x \leq x$.
If $x \leq y$ and $y \leq z$, then $y - x$ and $z - y$ are nonnegative. Their sum is nonnegative, and

$$(y - x) + (z - y) = z - x.$$

Therefore $x \leq z$.

# Antisymmetry

## Proposition

If $x \leq y$ and $y \leq x$, then $x = y$.

## Proof.

The two assumptions say that

$$y - x \in \mathbb{Q}_{\geq 0}, \qquad x - y \in \mathbb{Q}_{\geq 0}.$$

But $x - y = -(y - x)$. A rational and its negative can both be nonnegative only when that rational is zero. Hence $y - x = 0$, and therefore $x = y$. $\square$

# Totality

## Proposition

For all $x, y \in \mathbb{Q}$, either $x \leq y$ or $y \leq x$.

## Proof.

Apply the sign dichotomy to the gap $y - x$. Either

$$y - x \in \mathbb{Q}_{\geq 0},$$

which means $x \leq y$, or

$$-(y - x) = x - y \in \mathbb{Q}_{\geq 0},$$

which means $y \leq x$. $\square$

Together with reflexivity, transitivity, and antisymmetry, this makes $\mathbb{Q}$ a linearly ordered set.

# Translation preserves order

## Proposition

If $x \leq y$, then $x + t \leq y + t$ for every $t \in \mathbb{Q}$.

## Proof.

The relevant gap does not change:

$$(y + t) - (x + t) = y - x.$$

Since the right side is nonnegative, so is the left side. $\square$

Consequently, adding the same rational to both sides preserves and reflects both weak and strict inequalities.

# Products of nonnegative rationals

## Proposition

If $0 \leq x$ and $0 \leq y$, then $0 \leq xy$.

This is precisely closure of $\mathbb{Q}_{\geq 0}$ under multiplication, translated through the equivalence

$$0 \leq x \quad \Longleftrightarrow \quad x \in \mathbb{Q}_{\geq 0}.$$

Combining this fact with translation invariance yields the familiar rule: if $x \leq y$ and $0 \leq t$, then

$$xt \leq yt.$$

# The embeddings preserve and reflect order

## Proposition

For $p, q \in \mathbb{Z}$,

$$j(p) \leq j(q) \iff p \leq q.$$

Both fractions have denominator $1 > 0$, so the comparison criterion gives

$$\left[\frac{p}{1}\right] \leq \left[\frac{q}{1}\right] \iff p \cdot 1 \leq 1 \cdot q.$$

Since $i = j \circ \iota$, the embedding $i \colon \mathbb{N} \to \mathbb{Q}$ likewise preserves and reflects $\leq$ and $<$.

# Strictly positive products

## Proposition

If $0 < x$ and $0 < y$, then $0 < xy$.

Choose positive-denominator representatives

$$x = \left[\frac{a}{b}\right], \qquad y = \left[\frac{c}{d}\right].$$

Strict positivity means $a > 0$ and $c > 0$. Hence $ac > 0$, while $bd > 0$, so

$$xy = \left[\frac{ac}{bd}\right] > 0.$$

Thus multiplication by a positive rational strictly preserves order.

# An ordered field

We have now proved that $\mathbb{Q}$ is simultaneously

- a field;
- a linear order;
- ordered compatibly with addition;
- ordered compatibly with multiplication by nonnegative elements.

**Theorem**

The constructed rational numbers form a linearly ordered field.

Algebra and order are not separate structures here: both ultimately descend from the arithmetic and order of $\mathbb{Z}$.

# The Archimedean bound: reducing to integers

**Theorem**

For every $x \in \mathbb{Q}$, there exists $n \in \mathbb{N}$ such that $x \leq i(n)$.

If $x < 0$, choose $n = 0$. Otherwise choose a representation

$$x = \left[\frac{a}{b}\right]$$

with $a \geq 0$ and $b > 0$.
The Archimedean property of $\mathbb{Z}$ supplies $n \in \mathbb{N}$ with

$$a \leq \iota(n).$$

Since $b$ is a positive integer, $1 \leq b$, and since $\iota(n) \geq 0$,

$$\iota(n) \leq b\,\iota(n).$$

# The Archimedean bound: finishing the comparison

Combining the two integer inequalities gives

$$a \leq b\,\iota(n).$$

Both denominators below are positive, so the cross-multiplication criterion yields

$$\left[\frac{a}{b}\right] \leq \left[\frac{\iota(n)}{1}\right] = i(n).$$

**What the theorem rules out**

No rational number is larger than every natural number. The field $\mathbb{Q}$ contains arbitrarily large numbers.

# Final picture

We proved that

- rational numbers are equivalence classes of fractions, under the cross-multiplication criterion;
- $\mathbb{Q}$ is a field, and every element has the form $j(a)j(b)^{-1}$ with $a, b \in \mathbb{Z}$ and $b \neq 0$;
- $\mathbb{Q}$ is linearly ordered; addition preserves order, and products of nonnegative elements are nonnegative;
- the embeddings $\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q}$ preserve and reflect arithmetic and order;
- $\mathbb{Q}$ satisfies the Archimedean bound.

# Arithmetic in $\mathbb{N}$

# Why study arithmetic in $\mathbb{N}$?

The next construction, the real numbers, is motivated by an equation that cannot be solved in $\mathbb{Q}$:

$$x^2 = 2.$$

Proving this impossibility requires genuinely arithmetic tools, which this section develops:

1. Euclidean division, with a unique quotient and remainder;
2. the greatest common divisor and Bézout's identity;
3. Euclid's lemma: a prime dividing a product divides a factor;
4. the fundamental theorem of arithmetic: every number from 2 on is a product of primes in exactly one way.

Counting the prime factors on both sides of $A^2 = 2B^2$ will then produce the desired contradiction.

# Euclidean division

## Theorem

Let $n, d \in \mathbb{N}$ with $d > 0$. There are unique $q, r \in \mathbb{N}$ such that

$$\boxed{n = dq + r, \qquad r < d.}$$

The number $q$ is the quotient and $r$ is the remainder. The bound $r < d$ is what singles them out among all decompositions of $n$.

## Proof strategy

Keep the positive divisor $d$ fixed and use induction on the dividend $n$. Passing from $n$ to $S(n)$ either increments the remainder or resets it to zero and increments the quotient.

# Euclidean division: the induction

For $n = 0$, take $q = r = 0$. Then

$$0 = d \cdot 0 + 0, \qquad 0 < d.$$

Now suppose

$$n = dq + r, \qquad r < d.$$

Since $r < d$, the gap from $r$ to $d$ is positive: $d = r + S(y)$ for some $y \in \mathbb{N}$. Moving the successor through the sum gives $d = S(r) + y$, so $S(r) \leq d$. Hence there are two possibilities:

$$S(r) < d \qquad \text{or} \qquad S(r) = d.$$

These alternatives determine the quotient and remainder for $S(n)$.

# Euclidean division: the two cases

If $S(r) < d$, then
$$S(n) = S((dq + r)) = dq + S(r).$$
Thus the quotient stays $q$ and the new remainder is $S(r)$.
If $S(r) = d$, then
$$S(n) = dq + S(r) = dq + d = d(q + 1) + 0.$$
Thus the quotient becomes $q + 1$ and the new remainder is $0 < d$.
In either case $S(n)$ has the required decomposition, so induction proves existence for every $n \in \mathbb{N}$.

# Uniqueness of quotient and remainder

Suppose

$$n = dq + r = dq' + r', \qquad r, r' < d.$$

If $q < q'$, write $q' = q + s$ with $s > 0$. Writing $s = S(t)$ gives $ds = dt + d \geq d$, while distributivity gives $dq' = dq + ds$. Consequently,

$$n = dq + r < dq + d \leq dq + ds = dq' \leq dq' + r' = n,$$

a contradiction. The same argument rules out $q' < q$, so totality gives $q = q'$. Addition cancellation in $dq + r = dq + r'$ now gives $r = r'$. This proves uniqueness.

# Divisibility and prime numbers

For $a, b \in \mathbb{N}$, define

$$a \mid b \iff b = ac \text{ for some } c \in \mathbb{N}.$$

**Definition**

A number $p \in \mathbb{N}$ is prime if $2 \leq p$ and

$$d \mid p \implies d = 1 \text{ or } d = p.$$

Thus 0 and 1 are not prime, while $2, 3, 5, 7$ are prime. A number $n \geq 2$ that is not prime is composite; it can be written

$$n = ab \qquad \text{with} \qquad 2 \leq a < n, \qquad 2 \leq b < n.$$

# The greatest common divisor

Let $a, b \in \mathbb{N}$, not both zero.

**Definition**

The greatest common divisor of $a$ and $b$ is the natural number $g = \gcd(a, b)$ characterized by

1. $g \mid a$ and $g \mid b$;
2. if $d \mid a$ and $d \mid b$, then $d \mid g$.

Thus $g$ is a common divisor that is divisible by every common divisor. Two numbers with these properties divide each other and are therefore equal, so the two conditions determine $g$ uniquely; the Euclidean algorithm below produces one. Since divisibility respects the natural order for positive numbers, $g$ is also the largest common divisor in the usual order.

For example,

$$\gcd(84, 30) = 6.$$

# The Euclidean algorithm

Euclidean division can be repeated. For $a, b > 0$, begin with

$$\begin{aligned} a &= bq_0 + r_0, & 0 &\le r_0 < b, \\ b &= r_0 q_1 + r_1, & 0 &\le r_1 < r_0, \\ r_0 &= r_1 q_2 + r_2, & 0 &\le r_2 < r_1, \end{aligned}$$

and continue while the remainder is nonzero.
The remainders strictly decrease, and a strictly decreasing sequence of natural numbers must terminate. Each step preserves common divisors: since $a = bq_0 + r_0$, every common divisor of $b$ and $r_0$ divides $a$, and every common divisor of $a$ and $b$ divides $r_0 = a - bq_0$.

**Conclusion**

The last nonzero remainder is $\gcd(a, b)$, the greatest common divisor of $a$ and $b$.

# Bézout's identity

Each line of the Euclidean algorithm can be rearranged in $\mathbb{Z}$:

$$r_{i+1} = r_{i-1} - q_{i+1} r_i.$$

Starting with the last nonzero remainder and substituting backwards expresses it as an integer linear combination of the original inputs.

## Bézout's identity

For $a, b > 0$, there are $u, v \in \mathbb{Z}$ such that

$$\boxed{\gcd(a, b) = ua + vb.}$$

Here the natural numbers are viewed inside $\mathbb{Z}$.

In particular, if $\gcd(a, b) = 1$, then

$$ua + vb = 1$$

for suitable integers $u, v$.

# Euclid's lemma

## Lemma

If $p$ is prime and $p \mid ab$, then

$$\boxed{p \mid a \quad \text{or} \quad p \mid b.}$$

Suppose $p \nmid a$. Every common divisor of $p$ and $a$ divides the prime $p$, so it is 1 or $p$. The second possibility is excluded, hence $\gcd(p, a) = 1$.
Bézout's identity gives $up + va = 1$. Multiplying by $b$,

$$ubp + vab = b.$$

Both terms on the left are divisible by $p$: the first visibly, and the second because $p \mid ab$. Therefore $p \mid b$, proving the lemma.

# The fundamental theorem of arithmetic

**Theorem**

Every $n \in \mathbb{N}$ with $n \geq 2$ is a product of primes:

$$n = p_1 p_2 \cdots p_k.$$

This factorization is unique up to reordering the prime factors.

Equivalently, if

$$n = p_1 \cdots p_k = q_1 \cdots q_\ell$$

and both lists are arranged in nondecreasing order, then

$$k = \ell \qquad \text{and} \qquad p_i = q_i \quad \text{for every } i.$$

Existence comes from induction; uniqueness comes from Euclid's lemma.

# Existence of prime factorization

We use strong induction on $n$, which follows from the induction principle for $\mathbb{N}$.
If $n$ is prime, it is already a product consisting of one prime. If it is not prime, then

$$n = ab \qquad \text{with} \qquad 2 \leq a < n, \qquad 2 \leq b < n.$$

The induction hypothesis supplies prime factorizations

$$a = p_1 \cdots p_k, \qquad b = q_1 \cdots q_\ell.$$

Multiplying them gives

$$n = p_1 \cdots p_k q_1 \cdots q_\ell.$$

Thus every $n \geq 2$ has a prime factorization.

# Uniqueness of prime factorization

Suppose

$$p_1 \cdots p_k = q_1 \cdots q_\ell.$$

The prime $p_1$ divides the product on the right. Repeated application of Euclid's lemma shows that $p_1 \mid q_j$ for some $j$. Since $q_j$ is prime and $p_1 \neq 1$, this forces

$$p_1 = q_j.$$

Reorder the right-hand factors so that $q_j$ comes first and cancel this common nonzero factor.
Repeating the argument matches and cancels every prime on both sides.
The process cannot exhaust one side first: once every factor on one side is cancelled, the other side is a product of primes equal to 1, which is impossible because every prime is at least 2.
Therefore the two lists contain exactly the same primes with exactly the same multiplicities.
This proves uniqueness up to reordering.

# Prime multiplicities in a square

For a prime $p$ and a positive natural number $n$, let $\nu_p(n)$ be the number of times $p$ occurs in the unique prime factorization of $n$; the factorization of 1 is empty, so $\nu_p(1) = 0$.
Concatenating factorizations of $m$ and $n$ yields a factorization of $mn$; by uniqueness it is *the* factorization, so multiplicities add:

$$\nu_p(mn) = \nu_p(m) + \nu_p(n).$$

In particular,

$$\boxed{\nu_p(n^2) = 2\nu_p(n).}$$

Thus every prime occurs an even number of times in the factorization of a square. For the prime 2, this says that $\nu_2(n^2)$ is even.

# The equation $x^2 = 2$ in $\mathbb{Q}$

Suppose that $x \in \mathbb{Q}$ satisfied $x^2 = 2$. Replacing $x$ by $-x$ if necessary, we may assume $x \geq 0$. A positive-denominator representative then has the form

$$x = \left[\frac{\iota(A)}{\iota(B)}\right], \qquad A, B \in \mathbb{N}, \qquad B > 0.$$

Squaring and using the equality criterion for rational numbers gives

$$A^2 = 2B^2$$

in $\mathbb{N}$. Since $B > 0$, the right side is positive; hence $A \neq 0$ and both sides are positive, so their prime multiplicities are defined. Counting factors of 2 yields

$$2\nu_2(A) = \nu_2(A^2) = \nu_2(2B^2) = 1 + 2\nu_2(B).$$

The left side is even and the right side is odd, a contradiction. Hence $x^2 = 2$ has no solutions.

# Final picture

We proved that

- division with remainder is always possible, and the pair $(q, r)$ is unique;
- the Euclidean algorithm computes greatest common divisors and yields Bézout's identity;
- a prime dividing a product divides one of the factors;
- every natural number from 2 on factors into primes, uniquely up to reordering;
- prime multiplicities add on products, so every prime occurs an even number of times in a square;
- consequently, $x^2 = 2$ has no solution in $\mathbb{Q}$.

# The real numbers

# Why introduce the real numbers?

We proved that the equation

$$x^2 = 2$$

has no solution in $\mathbb{Q}$. Nevertheless, rational numbers can approximate the desired positive solution with arbitrarily small error.

We therefore seek a larger ordered field in which

- every rational number remains visible;
- limits of rational Cauchy sequences exist;
- arithmetic and order are compatible with taking limits.

**Construction**

A real number will be an equivalence class of Cauchy sequences of rational numbers. Two sequences represent the same real when their difference tends to zero.

# Absolute value

In any linearly ordered field, define

$$\boxed{|x| = \max\{x, -x\}.}$$

We shall use the following standard properties without reproving them.

## Order and symmetry

$$\begin{aligned} 0 &\le |x|, \\ |x| = 0 &\iff x = 0, \\ -|x| \le x &\le |x|, \\ |-x| &= |x|, \\ |x-y| &= |y-x|. \end{aligned}$$

For $\varepsilon \ge 0$,

$$|x| \le \varepsilon \iff -\varepsilon \le x \le \varepsilon.$$

## Arithmetic and estimates

$$\begin{aligned} |xy| &= |x|\,|y|, \\ |x^{-1}| &= |x|^{-1} \qquad (x \ne 0), \\ |x+y| &\le |x| + |y|, \\ \big||x| - |y|\big| &\le |x-y|. \end{aligned}$$

For $\delta \ge 0$,

$$\delta < |x| \iff x < -\delta \quad \text{or} \quad \delta < x.$$

# Cauchy sequences and pre-reals

Let $x = (x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{Q}$.

## Definition

The sequence $x$ is Cauchy if

$$\forall\varepsilon > 0\ \exists N\ \forall p, q \geq N, \qquad |x_p - x_q| \leq \varepsilon.$$

Here $\varepsilon$ is rational.

The condition asks the tail of the sequence to have arbitrarily small diameter. It does not mention a limit, since that limit may not lie in $\mathbb{Q}$.

## Definition

A pre-real is a rational Cauchy sequence. We write

$$\mathbb{R}_{\mathrm{pre}} = \{x \colon \mathbb{N} \to \mathbb{Q} : x \text{ is Cauchy}\}.$$

## Every Cauchy sequence is bounded: the finite part

Let $x$ be Cauchy. Apply the definition with $\varepsilon = 1$. There is $A \in \mathbb{N}$ such that

$$p, q \geq A \quad \Longrightarrow \quad |x_p - x_q| \leq 1.$$

The initial set

$$\{|x_0|, |x_1|, \ldots, |x_A|\}$$

is finite and nonempty, so it has a maximum $M$. In particular,

$$0 \leq M, \qquad |x_n| \leq M \quad (0 \leq n \leq A).$$

We claim that $B = M + 1$ bounds the whole sequence.

# Every Cauchy sequence is bounded: the tail

If $n \leq A$, then

$$|x_n| \leq M \leq M + 1.$$

If $A \leq n$, the triangle inequality and the Cauchy estimate give

$$|x_n| = |x_A + (x_n - x_A)| \leq |x_A| + |x_n - x_A| \leq M + 1.$$

Thus

$$\boxed{B = M + 1 > 0, \qquad |x_n| \leq B \text{ for every } n.}$$

This boundedness lemma is the key extra ingredient in proving that products and reciprocals of suitable Cauchy sequences are again Cauchy.

# When do two pre-reals represent the same number?

## Definition

For $x, y \in \mathbb{R}_{\mathrm{pre}}$, define

$$x \sim y \quad \Longleftrightarrow \quad \forall \varepsilon > 0\ \exists N\ \forall n \geq N, \qquad |x_n - y_n| \leq \varepsilon.$$

In words, $x \sim y$ means that $x_n - y_n$ tends to zero.
Reflexivity follows from $|x_n - x_n| = 0$. Symmetry follows from

$$|y_n - x_n| = |x_n - y_n|.$$

Transitivity is the only part requiring an estimate.

# Transitivity of the equivalence relation

Suppose $x \sim y$ and $y \sim z$, and fix $\varepsilon > 0$. Choose thresholds $N, M$ such that

$$|x_n - y_n| \leq \frac{\varepsilon}{2} \quad (n \geq N), \qquad |y_n - z_n| \leq \frac{\varepsilon}{2} \quad (n \geq M).$$

For $n \geq \max(N, M)$,

$$\begin{aligned} |x_n - z_n| &= |(x_n - y_n) + (y_n - z_n)| \\ &\leq |x_n - y_n| + |y_n - z_n| \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Hence $x \sim z$, and $\sim$ is an equivalence relation on the pre-reals.

# Constants, negation, and addition

Define operations pointwise:

$$0_n = 0, \qquad 1_n = 1, \qquad (-x)_n = -x_n, \qquad (x+y)_n = x_n + y_n.$$

Constant sequences are Cauchy, and negation preserves every Cauchy estimate because

$$|(-x_p) - (-x_q)| = |x_q - x_p| = |x_p - x_q|.$$

For addition, choose $N, M$ so that the variations of $x$ and $y$ are at most $\varepsilon/2$. If $p, q \geq \max(N, M)$, then

$$\begin{aligned} |(x_p + y_p) - (x_q + y_q)| &\leq |x_p - x_q| + |y_p - y_q| \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus $-x$ and $x + y$ are pre-reals.

# Addition respects equivalence

Suppose $x \sim x'$ and $y \sim y'$. Given $\varepsilon > 0$, choose a common threshold after which

$$|x_n - x_n'| \le \frac{\varepsilon}{2}, \qquad |y_n - y_n'| \le \frac{\varepsilon}{2}.$$

Beyond that threshold,

$$\begin{aligned} |(x_n + y_n) - (x_n' + y_n')| &= |(x_n - x_n') + (y_n - y_n')| \\ &\le |x_n - x_n'| + |y_n - y_n'| \\ &\le \varepsilon. \end{aligned}$$

Therefore

$$x + y \sim x' + y'.$$

The same argument, using symmetry of absolute value, shows that $x \sim x'$ implies $-x \sim -x'$.

# Products of Cauchy sequences

Let $|x_n| \leq A$ and $|y_n| \leq B$, with $A, B > 0$. Choose a common threshold such that

$$|x_p - x_q| \leq \frac{\varepsilon}{2B}, \qquad |y_p - y_q| \leq \frac{\varepsilon}{2A}.$$

Use the identity

$$x_p y_p - x_q y_q = x_p(y_p - y_q) + y_q(x_p - x_q).$$

Hence

$$\begin{aligned} |x_p y_p - x_q y_q| &\leq |x_p|\,|y_p - y_q| + |y_q|\,|x_p - x_q| \\ &\leq A\frac{\varepsilon}{2A} + B\frac{\varepsilon}{2B} = \varepsilon. \end{aligned}$$

Thus the pointwise product $(xy)_n = x_n y_n$ is Cauchy.

# Multiplication respects equivalence

Suppose $x \sim x'$ and $y \sim y'$. Choose bounds

$$|x_n| \leq A, \qquad |y'_n| \leq B, \qquad A, B > 0.$$

The comparison of the two products is organized as

$$x_n y_n - x'_n y'_n = x_n(y_n - y'_n) + y'_n(x_n - x'_n).$$

Once

$$|y_n - y'_n| \leq \frac{\varepsilon}{2A}, \qquad |x_n - x'_n| \leq \frac{\varepsilon}{2B},$$

we obtain

$$|x_n y_n - x'_n y'_n| \leq A\frac{\varepsilon}{2A} + B\frac{\varepsilon}{2B} = \varepsilon.$$

Therefore $xy \sim x'y'$.

# A nonzero pre-real stays away from zero

The condition $x \not\sim 0$ means that for some $\delta > 0$, no tail satisfies $|x_n| \leq \delta$. Thus

$$\exists \delta > 0 \; \forall N \; \exists M \geq N, \qquad \delta < |x_M|.$$

Apply the Cauchy property with $\delta/2$, obtaining a threshold $N$. Choose $M \geq N$ with $|x_M| > \delta$. For every $n \geq M$,

$$\begin{aligned} |x_n| &= |x_M - (x_M - x_n)| \\ &\geq |x_M| - |x_M - x_n| \\ &> \delta - \frac{\delta}{2} = \frac{\delta}{2}. \end{aligned}$$

Hence there are $A > 0$ and $M$ such that

$$\boxed{A < |x_n| \quad \text{for every } n \geq M.}$$

# Reciprocals of nonzero pre-reals are Cauchy

Let $x \not\sim 0$. By the preceding lemma, choose $A > 0$ and $N$ with $A < |x_n|$ for $n \geq N$. Also choose $M$ so that

$$|x_p - x_q| \leq \varepsilon A^2 \qquad (p, q \geq M).$$

For $p, q \geq \max(N, M)$, both denominators are nonzero and

$$\begin{aligned} \left| \frac{1}{x_p} - \frac{1}{x_q} \right| &= \frac{|x_q - x_p|}{|x_p|\,|x_q|} \\ &\leq \frac{|x_q - x_p|}{A^2} \\ &\leq \frac{\varepsilon A^2}{A^2} = \varepsilon. \end{aligned}$$

Thus $(1/x_n)$ is Cauchy whenever the class of $x$ is nonzero.

# The reciprocal is independent of representatives

Suppose $x \sim x'$, with neither sequence equivalent to zero. Choose eventual lower bounds

$$A < |x_n|, \qquad A' < |x'_n|.$$

Since $x \sim x'$, eventually

$$|x_n - x'_n| \le \varepsilon AA'.$$

Beyond a common threshold,

$$\begin{aligned} \left| \frac{1}{x_n} - \frac{1}{x'_n} \right| &= \frac{|x'_n - x_n|}{|x_n|\,|x'_n|} \\ &\le \frac{|x'_n - x_n|}{AA'} \\ &\le \varepsilon. \end{aligned}$$

Therefore $(1/x_n) \sim (1/x'_n)$. Reciprocal is consequently well defined on every nonzero equivalence class.

# Definition of the real numbers

We have proved that $\sim$ is an equivalence relation on $\mathbb{R}_{\text{pre}}$.

## Definition

The real numbers are the quotient

$$\boxed{\mathbb{R} = \mathbb{R}_{\text{pre}}/\sim.}$$

We write $X = [x]$ when the real number $X$ is represented by the Cauchy sequence $x$.

If $X = [x]$ and $Y = [y]$, the operations descend from pointwise operations:

$$X + Y = [x + y], \qquad -X = [-x], \qquad XY = [xy].$$

If $X = [x] \neq 0$, choose $N$ with $x_n \neq 0$ for $n \geq N$. Its reciprocal is represented by any rational sequence whose tail is $(1/x_n)_{n \geq N}$; the finitely many earlier values do not affect its class. The estimates already proved show that every formula is independent of representatives.

# The commutative-ring laws

Write $X = [x]$, $Y = [y]$, and $Z = [z]$. Every ring law holds pointwise before passing to the quotient. For example,

$$((x + y) + z)_n = (x_n + y_n) + z_n = x_n + (y_n + z_n) = (x + (y + z))_n,$$

so

$$(X + Y) + Z = X + (Y + Z).$$

The same representative calculation proves

- associativity and commutativity of addition and multiplication;
- the zero and one identity laws;
- additive inverses;
- distributivity.

Thus $\mathbb{R}$ is a commutative ring.

# The real numbers are nontrivial

The zero and one of $\mathbb{R}$ are represented by the constant sequences

$$(0,0,0,\ldots), \qquad (1,1,1,\ldots).$$

If these classes were equal, then for $\varepsilon = 1/2$ there would be a threshold $N$ such that

$$|0-1| \le \frac{1}{2} \qquad (n \ge N).$$

But $|0-1| = 1$, a contradiction. Therefore

$$\boxed{0 \ne 1 \text{ in } \mathbb{R}.}$$

Hence the quotient contains at least two distinct elements.

# Multiplying by a reciprocal

Let $X = [x] \neq 0$. Then $x \not\sim 0$, so there are $\delta > 0$ and $N$ such that

$$\delta < |x_n| \qquad (n \geq N).$$

In particular, $x_n \neq 0$ for $n \geq N$. On this tail,

$$x_n \frac{1}{x_n} = 1.$$

Hence the product sequence differs from the constant sequence 1 only in finitely many positions. Their difference is eventually zero, so they are equivalent:

$$\boxed{XX^{-1} = 1 \qquad (X \neq 0).}$$

Therefore the constructed real numbers form a field.

# Embedding the rational numbers

A rational number determines a constant Cauchy sequence. Define

$$k\colon \mathbb{Q} \longrightarrow \mathbb{R}, \qquad k(r) = [(r, r, r, \dots)].$$

Pointwise arithmetic gives

$$\begin{gathered} k(0) = 0, \qquad k(1) = 1, \qquad k(-r) = -k(r), \\ k(r+s) = k(r) + k(s), \qquad k(rs) = k(r)k(s). \end{gathered}$$

Thus $k$ is a field homomorphism. It remains to check that distinct rationals do not become equivalent as constant sequences.

# The rational embedding is injective

Suppose $k(r) = k(s)$. Then the two constant sequences are equivalent:

$$\forall \varepsilon > 0 \; \exists N \; \forall n \geq N, \qquad |r - s| \leq \varepsilon.$$

If $r \neq s$, then $|r - s| > 0$. Choose

$$\varepsilon = \frac{|r - s|}{2} > 0.$$

Equivalence would give

$$|r - s| \leq \frac{|r - s|}{2},$$

which is impossible. Hence $r = s$, so $k$ is injective.
We may therefore regard $\mathbb{Q}$ as a subfield of $\mathbb{R}$.

# A positive pre-real

## Definition

A pre-real $x$ is positive, written $\mathrm{Pos}(x)$, if

$$\exists \delta > 0 \; \exists N \; \forall n \geq N, \qquad \delta \leq x_n.$$

A positive pre-real is not equivalent to zero. Indeed, if $x \sim 0$, fix the positive rational $\delta/2$. At a common large index we would have

$$\delta \leq x_n \qquad \text{and} \qquad |x_n| \leq \frac{\delta}{2},$$

which is impossible.

# Why term-by-term sign conditions do not work

One might try to declare

$$x > 0 \iff x_n > 0 \text{ for every } n, \qquad x \geq 0 \iff x_n \geq 0 \text{ for every } n.$$

Neither condition gives a well-defined property of an equivalence class.

1. *Changing finitely many terms changes the answer.* The equivalent pre-reals

   $$(1, 1, 1, \ldots) \sim (-1, 1, 1, \ldots)$$

   do not both have all terms positive. Likewise, $(0, 0, 0, \ldots) \sim (-1, 0, 0, \ldots)$, but only the first has all terms nonnegative.

2. *Even requiring the sign only eventually is not enough.* Set $u_n = 1/(n + 1)$. Then every $u_n > 0$, but $u \sim 0$; the naive rule would therefore make the zero class positive. Also $-u \sim 0$, although every term of $-u$ is negative, so termwise nonnegativity would depend on the chosen representative of zero.

A fixed eventual gap separates positive classes from zero. Nonnegative will mean: equivalent to zero or positive.

# Positivity is independent of representatives

Suppose $x \sim x'$ and $x_n \geq \delta > 0$ for $n \geq N$. Choose $M$ such that

$$|x_n - x_n'| \leq \frac{\delta}{2} \qquad (n \geq M).$$

For $n \geq \max(N, M)$,

$$\begin{aligned} x_n' &= x_n - (x_n - x_n') \\ &\geq \delta - |x_n - x_n'| \\ &\geq \delta - \frac{\delta}{2} = \frac{\delta}{2}. \end{aligned}$$

Hence $\mathrm{Pos}(x')$. By symmetry, positivity depends only on the equivalence class represented by the sequence.

# Positive pre-reals are closed under arithmetic

Suppose eventually

$$x_n \geq A > 0, \qquad y_n \geq B > 0.$$

Beyond the maximum of the two thresholds,

$$x_n + y_n \geq A + B > 0,$$

so $x + y$ is positive.
Since all four quantities are nonnegative,

$$x_n y_n \geq AB > 0,$$

so $xy$ is positive as well.

## Conclusion

$$\operatorname{Pos}(x), \operatorname{Pos}(y) \implies \operatorname{Pos}(x+y), \operatorname{Pos}(xy).$$

# Nonnegative pre-reals

## Definition

A pre-real $x$ is nonnegative if

$$\mathrm{Nonneg}(x) \iff \mathrm{Pos}(x) \ \text{ or } \ x \sim 0.$$

This property is invariant under equivalence because both alternatives are. It includes every eventually pointwise nonnegative pre-real. Indeed, if $x_n \geq 0$ for every $n \geq N$, then:

- if $x \sim 0$, it is nonnegative by definition;
- otherwise, $|x_n| > \delta$ eventually for some $\delta > 0$, and the pointwise sign forces $x_n > \delta$.

Thus $x$ is positive in the second case.

# Opposite nonnegative signs meet only at zero

Suppose both $x$ and $-x$ are nonnegative. If $x \not\sim 0$, then also $-x \not\sim 0$, so both must be positive. There would be $\delta, \delta' > 0$ and a common large index $n$ with

$$x_n \geq \delta, \qquad -x_n \geq \delta'.$$

Adding yields

$$0 \geq \delta + \delta' > 0,$$

a contradiction. Therefore

$$\boxed{\text{Nonneg}(x), \text{Nonneg}(-x) \implies x \sim 0.}$$

This is the antisymmetry mechanism for the real order.

# If a pre-real is not nonnegative

Assume $x$ is not nonnegative. Then $x \not\sim 0$ and $x$ is not positive.
From $x \not\sim 0$, choose $\delta > 0$ and $N$ such that

$$|x_n| > \delta \qquad (n \geq N).$$

From the Cauchy property, choose $M$ such that

$$|x_p - x_q| \leq \frac{\delta}{2} \qquad (p, q \geq M).$$

Since $x$ is not positive, the proposed gap $\delta$ fails after every threshold. Hence there is $R \geq \max(N, M)$ with $x_R < \delta$. Together with $|x_R| > \delta$, this forces

$$x_R < -\delta, \qquad \text{so} \qquad -x_R > \delta.$$

# The opposite pre-real is positive

Continue with the index $R$ for which $-x_R > \delta$. If $n \geq \max(R, N, M)$, the Cauchy estimate gives

$$x_R - x_n \geq -\frac{\delta}{2}.$$

Therefore

$$\begin{aligned} -x_n &= (x_R - x_n) + (-x_R) \\ &> -\frac{\delta}{2} + \delta = \frac{\delta}{2}. \end{aligned}$$

Thus $-x$ is positive, and in particular nonnegative. We have proved the sign alternative

$$\boxed{\mathrm{Nonneg}(x) \quad \text{or} \quad \mathrm{Nonneg}(-x).}$$

This is the key step toward totality of the order.

# Definition of the real order

Nonnegativity passes to equivalence classes. For $X, Y \in \mathbb{R}$, define

$$\boxed{X \leq Y \quad \Longleftrightarrow \quad Y - X \text{ is nonnegative.}}$$

Reflexivity follows because $X - X = 0$. For transitivity, if $Y - X$ and $Z - Y$ are nonnegative, their sum is nonnegative, and

$$(Z - Y) + (Y - X) = Z - X.$$

For antisymmetry, if $X \leq Y$ and $Y \leq X$, then both $Y - X$ and $-(Y - X) = X - Y$ are nonnegative. The preceding lemma gives

$$Y - X = 0,$$

hence $X = Y$.

# Totality and compatibility with arithmetic

Apply the sign alternative to $Y - X$. If $Y - X$ is nonnegative, then $X \leq Y$. Otherwise $-(Y - X) = X - Y$ is nonnegative, so $Y \leq X$. Thus the order is total.
Translation preserves order because

$$(Y + T) - (X + T) = Y - X.$$

Products of nonnegative classes are nonnegative by the corresponding pre-real result. Products of positive classes are positive.

**Theorem**

The constructed real numbers form a linearly ordered field.

# Eventual inequalities are sufficient

The eventual non-strict sign condition is not necessary, but it is a useful sufficient condition:

$$\boxed{x_n \geq 0 \text{ eventually} \quad \Longrightarrow \quad [x] \geq 0.}$$

Indeed, if $x \sim 0$, then $[x] = 0$. Otherwise, $|x_n| > \delta$ eventually for some $\delta > 0$; combined with $x_n \geq 0$, this gives $x_n > \delta$ eventually, so $x$ is positive.
Consequently, if $X = [x]$ and $Y = [y]$, then

$$\boxed{x_n \leq y_n \text{ eventually} \quad \Longrightarrow \quad X \leq Y,}$$

because $y_n - x_n \geq 0$ eventually.
The converses fail. For example, $z_n = -1/(n+1)$ satisfies $z \sim 0$, so $[z] = 0 \geq 0$, although no term of $z$ is nonnegative. Likewise, if $x_n = 0$ and $y_n = z_n$, then $[x] = [y]$, but $x_n \leq y_n$ never holds.

# The rational embedding preserves order

For $r, s \in \mathbb{Q}$, the difference $k(s) - k(r)$ is represented by the constant sequence $s - r$. If $r < s$, this sequence is positive with the fixed gap $s - r$. If $r = s$, it represents zero. Hence

$$r \leq s \quad \Longrightarrow \quad k(r) \leq k(s).$$

Conversely, if $s - r < 0$, the constant sequence $s - r$ is neither positive nor equivalent to zero, so it cannot be nonnegative. Therefore

$$\boxed{k(r) \leq k(s) \quad \Longleftrightarrow \quad r \leq s.}$$

Injectivity then also gives $k(r) < k(s) \Longleftrightarrow r < s$.

# A Cauchy tail controls the real distance

Let $X = [x] \in \mathbb{R}$, let $\varepsilon \in \mathbb{Q}$ be positive, and suppose $N$ works in the Cauchy condition for $x$ at tolerance $\varepsilon$:

$$p, q \geq N \quad \Longrightarrow \quad |x_p - x_q| \leq \varepsilon.$$

Fixing $q = N$, we obtain

$$-\varepsilon \leq x_n - x_N \leq \varepsilon \qquad (n \geq N).$$

Now $k(x_N)$ is represented by the constant sequence $(x_N, x_N, \ldots)$. Therefore

$$X - k(x_N) = [(x_n - x_N)_{n \in \mathbb{N}}].$$

The eventual inequalities pass to the real order, giving

$$-k(\varepsilon) \leq X - k(x_N) \leq k(\varepsilon)$$

hence

$$\boxed{|X - k(x_N)| \leq k(\varepsilon).}$$

# A positive rational below a positive real

Suppose $X = [x] > 0$. Positivity of its representative gives $\delta > 0$ and $N$ such that

$$x_n \geq \delta \qquad (n \geq N).$$

Set $r = \delta/2$. Then $r > 0$, and eventually

$$x_n - r \geq \frac{\delta}{2} > 0.$$

Hence

$$0 < k(r) < X.$$

(The positive gap $\delta/2$ makes $k(r) < X$ strict.)

# The rationals are dense among the reals

Let $X \in \mathbb{R}$, and let $E > 0$ be real. The previous lemma, applied to $E$, provides a rational $\varepsilon > 0$ with

$$0 < k(\varepsilon) < E.$$

The Cauchy-tail estimate at tolerance $\varepsilon$ then provides $r \in \mathbb{Q}$ with

$$|X - k(r)| \leq k(\varepsilon) < E.$$

Thus every real number can be approximated by a rational number with arbitrarily small real error:

$$\boxed{\forall E > 0\ \exists r \in \mathbb{Q}, \qquad |X - k(r)| < E.}$$

# The Archimedean bound in $\mathbb{R}$

Let $X = [x]$. Boundedness of the pre-real $x$ gives a positive rational $B$ such that

$$|x_m| \leq B \qquad \text{for every } m.$$

*Convention.* From now on we write $n$ for the rational number $i(n)$ whenever a natural number appears inside a rational or real expression; carrying the embeddings explicitly would only clutter the estimates.

The Archimedean property of $\mathbb{Q}$ gives $n \in \mathbb{N}$ with $B \leq n + 1$. Therefore

$$x_m \leq |x_m| \leq B \leq n + 1$$

for every $m$. Passing to classes gives

$$\boxed{X \leq k(n+1).}$$

Thus the real construction inherits an Archimedean upper bound directly from the boundedness of rational Cauchy sequences.

# Arbitrarily small reciprocal bounds

Let $E > 0$. Apply the Archimedean bound to $|E^{-1}|$. For some $n \in \mathbb{N}$,

$$|E^{-1}| \leq k(n+1).$$

Since $E > 0$, its reciprocal is positive, so the left side equals $E^{-1}$. Both sides are positive; reversing the inequality under reciprocals yields

$$\boxed{k\left(\frac{1}{n+1}\right) \leq E.}$$

This is the form of the Archimedean property used in both Cauchy estimates below: the rational numbers $1/(n+1)$ eventually fit below every positive real number.

# Convergence and Cauchy sequences in $\mathbb{R}$

For a sequence $F\colon \mathbb{N} \to \mathbb{R}$, define

$$F_n \longrightarrow X$$

to mean

$$\forall E > 0\ \exists N\ \forall n \geq N, \qquad |F_n - X| \leq E.$$

The sequence is Cauchy if

$$\forall E > 0\ \exists N\ \forall p, q \geq N, \qquad |F_p - F_q| \leq E.$$

The real numbers are complete if every real Cauchy sequence converges to a real number. The construction was designed so that a rational Cauchy sequence converges to the class it represents.

# Rational tolerances are enough

Let $F\colon \mathbb{N} \to \mathbb{R}$ and $X \in \mathbb{R}$. Suppose that for every positive rational $\varepsilon$ we can prove

$$\exists N \; \forall n \geq N, \qquad |F_n - X| \leq k(\varepsilon).$$

Then $F_n \to X$ for arbitrary positive real tolerances.
Indeed, given $E > 0$, choose a positive rational $\delta$ with

$$0 < k(\delta) < E.$$

The hypothesis at $\delta$ supplies $N$ such that, for $n \geq N$,

$$|F_n - X| \leq k(\delta) < E,$$

and in particular $|F_n - X| \leq E$.
We will repeatedly use this criterion below.

# A representative sequence converges: the idea

Let $x = (x_n)_{n \in \mathbb{N}}$ represent $X = [x]$. Applying the rational embedding term by term gives the real sequence

$$k(x_0),\ k(x_1),\ k(x_2), \dots$$

We claim that this sequence converges to $X$. Each $k(x_n)$ is represented by the constant row obtained by *freezing* $x_n$, whereas $X$ is represented by the moving bottom row:

| real number | a rational representative | | | | |
|---|---|---|---|---|---|
| $k(x_0)$ | $x_0$ | $x_0$ | $x_0$ | $x_0$ | $\cdots$ |
| $k(x_1)$ | $x_1$ | $x_1$ | $x_1$ | $x_1$ | $\cdots$ |
| $k(x_2)$ | $x_2$ | $x_2$ | $x_2$ | $x_2$ | $\cdots$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | |
| $k(x_n)$ | $x_n$ | $x_n$ | $x_n$ | $x_n$ | $\cdots$ |
| $X = [x]$ | $x_0$ | $x_1$ | $x_2$ | $x_3$ | $\cdots$ |

Thus $k(x_n) \to X$ means that, as we freeze terms farther along the sequence, the constant rows get closer to the moving bottom row. The Cauchy property says exactly that late terms stay close to one another, and that is all the claim requires.

# A representative sequence converges to its class

Let $x \in \mathbb{R}_{\text{pre}}$ and set $X = [x]$. We claim

$$k(x_n) \longrightarrow X.$$

Fix a positive rational $\varepsilon$. The Cauchy property supplies $N$ such that for $n, m \geq N$,

$$|x_n - x_m| \leq \varepsilon.$$

In the table, this says that beyond column $N$ every frozen row with $n \geq N$ agrees with the moving row to within $\varepsilon$; the finitely many earlier columns are irrelevant, because the quotient order only compares tails. As explained above, this gives

$$|k(x_n) - X| \leq k(\varepsilon) \qquad (n \geq N).$$

We have verified the rational-tolerance hypothesis for every $\varepsilon > 0$ in $\mathbb{Q}$. The preceding reduction therefore gives

$$\boxed{k(x_n) \longrightarrow X.}$$

# Completeness: statement and strategy

**Theorem**

Every Cauchy sequence of real numbers converges to a real number.

Let $F : \mathbb{N} \to \mathbb{R}$ be a Cauchy sequence. The proof has three steps.

1. Approximate. Rational density provides, for each $n$, a rational $x_n$ with $\left|F_n - k(x_n)\right| < k\left(\frac{1}{n+1}\right)$.
2. Transfer the Cauchy property. The rational sequence $x = (x_n)$ is Cauchy because $F$ is; it is therefore a pre-real and defines a real number $X = [x]$.
3. Pass to the limit. We already know that $k(x_n) \to X$, and the approximation errors tend to zero; hence $F_n \to X$.

# Approximating a real sequence term by term

Let $F\colon \mathbb{N} \to \mathbb{R}$. Rational density gives, for every $n$, a rational $x_n$ satisfying

$$\boxed{|F_n - k(x_n)| < k\left(\frac{1}{n+1}\right).}$$

Choosing one such rational for each $n$ defines a sequence

$$x = \mathrm{approx}(F)\colon \mathbb{N} \longrightarrow \mathbb{Q}.$$

Any tolerances shrinking to zero would serve; the choice $1/(n+1)$ guarantees that the error at stage $n$ tends to zero, so $F_n$ and $k(x_n)$ will have the same limiting behaviour.
The goal is to prove that $F_n$ converges to the real number represented by $x$. To speak of that number at all, we must first prove that $x$ is a rational Cauchy sequence whenever $F$ is a real Cauchy sequence.

# The rational approximation is Cauchy

Assume $F$ is Cauchy, and fix a positive rational $\varepsilon$. Choose $N$ such that

$$|F_p - F_q| \leq \frac{k(\varepsilon)}{3} \qquad (p, q \geq N),$$

and use the Archimedean property to choose $M$ such that

$$k\left(\frac{1}{M+1}\right) \leq \frac{k(\varepsilon)}{3}.$$

For $p \geq M$, we have $M+1 \leq p+1$; taking reciprocals reverses this comparison, so the approximation bound gives

$$|F_p - k(x_p)| < k\left(\frac{1}{p+1}\right) \leq k\left(\frac{1}{M+1}\right) \leq \frac{k(\varepsilon)}{3}.$$

If $p, q \geq \max(M, N)$, splitting $k(x_p) - k(x_q)$ into three differences gives

$$\begin{aligned} |k(x_p) - k(x_q)| &\leq |k(x_p) - F_p| + |F_p - F_q| + |F_q - k(x_q)| \\ &\leq \frac{k(\varepsilon)}{3} + \frac{k(\varepsilon)}{3} + \frac{k(\varepsilon)}{3} = k(\varepsilon). \end{aligned}$$

Since $k$ reflects order, $|x_p - x_q| \leq \varepsilon$. Thus $x$ is a rational Cauchy sequence.

# Every real Cauchy sequence converges

The rational approximation $x$ is a pre-real, so let

$$X = [x] \in \mathbb{R}.$$

We already know that $k(x_n) \to X$.
Fix a positive real $\varepsilon$. Choose $N$ so that

$$|k(x_n) - X| \leq \frac{\varepsilon}{2} \qquad (n \geq N),$$

and use the Archimedean property to choose $M$ so that

$$k\left(\frac{1}{M+1}\right) \leq \frac{\varepsilon}{2}.$$

For $n \geq M$, reciprocals reverse the comparison $M+1 \leq n+1$, so the approximation bound gives

$$|F_n - k(x_n)| < k\left(\frac{1}{n+1}\right) \leq k\left(\frac{1}{M+1}\right) \leq \frac{\varepsilon}{2}.$$

For $n \geq \max(N, M)$, combining the two halves,

$$\begin{aligned} |F_n - X| &\leq |F_n - k(x_n)| + |k(x_n) - X| \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Hence $F_n \to X$.

# The real numbers are complete

**Theorem**

Every Cauchy sequence of real numbers converges to a real number.

Looking back at the three steps:

- rational density produced the approximations $x_n$, with errors controlled by $1/(n+1)$;
- the Cauchy hypothesis on $F$ was used exactly once: to prove that the rational approximation $x$ is itself Cauchy;
- the convergence $k(x_n) \to X$ came directly from the construction of $\mathbb{R}$, and the vanishing errors transferred it to $F$.

The construction has repaired the defect that motivated it: Cauchy sequences that had no limit in $\mathbb{Q}$ acquire one in $\mathbb{R}$.

# Newton's rational approximations

Define a rational sequence $a = (a_n)$ recursively by

$$a_0 = 1, \qquad a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right).$$

The idea: if $a_n^2$ falls short of 2, then $(2/a_n)^2$ overshoots it, and vice versa, so the average of $a_n$ and $2/a_n$ should approximate the desired number better than either.

The sequence is well defined and positive. Indeed, $a_0 > 0$, and if $a_n > 0$, then both $a_n$ and $2/a_n$ are positive, so their average is positive. The first few terms are

$$1, \qquad \frac{3}{2}, \qquad \frac{17}{12}, \qquad \frac{577}{408}, \quad \ldots$$

## The plan

Show that the errors $a_n^2 - 2$ tend to zero, deduce that $(a_n)$ is Cauchy, and prove that the real number it represents squares to 2.

# The exact error formula

Define the rational error sequence $e = (e_n)$ by

$$e_n = a_n^2 - 2.$$

Direct calculation gives

$$\begin{aligned} e_{n+1} &= \frac{1}{4}\left(a_n + \frac{2}{a_n}\right)^2 - 2 \\ &= \frac{a_n^2 + 4 + 4/a_n^2 - 8}{4} \\ &= \boxed{\frac{(a_n^2 - 2)^2}{4a_n^2}} = \frac{e_n^2}{4a_n^2}. \end{aligned}$$

The right side is a square divided by a positive number, so $e_{n+1} \geq 0$ for every $n$: from index 1 on, the approximations overshoot,

$$a_n^2 \geq 2 \qquad (n \geq 1).$$

Also $a_1 = 3/2$, hence $e_1 = 1/4$.

# The errors tend to zero

For $n \geq 1$, we have $a_n^2 \geq 2$, so

$$0 \leq e_{n+1} = \frac{e_n^2}{4a_n^2} \leq \frac{e_n^2}{8}.$$

Starting from $e_1 = 1/4$, induction gives $e_n \leq 1/4$. Consequently,

$$e_{n+1} \leq \frac{e_n}{32},$$

and another induction yields

$$\boxed{0 \leq e_n \leq \frac{1}{4 \cdot 32^{n-1}} \qquad (n \geq 1).}$$

These upper bounds tend to zero. Given a rational $\varepsilon > 0$, the Archimedean property provides $m$ with $m + 1 \geq 1/(4\varepsilon)$, and $32^m \geq m + 1$ by induction, so

$$e_n \leq \frac{1}{4 \cdot 32^m} \leq \frac{1}{4(m+1)} \leq \varepsilon \qquad (n \geq m+1).$$

# The Newton sequence is Cauchy

Since $a_n > 0$ and $a_n^2 \geq 2$, we have $a_n \geq 1$ for $n \geq 1$: a number below 1 would have square below 1. Hence, from index 1 on, the sequence decreases by a controlled amount:

$$0 \leq a_n - a_{n+1} = \frac{a_n^2 - 2}{2a_n} = \frac{e_n}{2a_n} \leq \frac{e_n}{2}.$$

If $q > p \geq 1$, the differences telescope, and iterating $e_{j+1} \leq e_j/32$ bounds them by a geometric series:

$$\begin{aligned} |a_p - a_q| &= \sum_{j=p}^{q-1} (a_j - a_{j+1}) \leq \frac{1}{2} \sum_{j=p}^{q-1} e_j \\ &\leq \frac{e_p}{2} \sum_{i=0}^{q-p-1} \frac{1}{32^i} < \frac{16}{31} e_p < e_p. \end{aligned}$$

Given a rational $\varepsilon > 0$, choose $N \geq 1$ with $e_N \leq \varepsilon$. The errors decrease, so for $q > p \geq N$,

$$|a_p - a_q| < e_p \leq e_N \leq \varepsilon.$$

Thus $(a_n)$ is Cauchy.

# A real number whose square is 2

The rational sequence $a = (a_n)$ is a pre-real, so define

$$\alpha = [a] \in \mathbb{R}.$$

Since $a_n \geq 1$, the sequence has a fixed positive gap and $\alpha > 0$.
Pointwise multiplication gives $\alpha^2 = [(a_n^2)]$, so

$$\alpha^2 - 2 = [e].$$

Given a rational $\varepsilon > 0$, the error estimate supplies $N \geq 1$ such that for $n \geq N$,

$$|e_n| = e_n \leq \varepsilon,$$

the errors being nonnegative from index 1 on. Thus $e \sim 0$, and therefore

$$\boxed{\alpha^2 = 2.}$$

No rational has this property, so $\alpha$ is a genuinely new number in the Cauchy completion of $\mathbb{Q}$.

# The equation $x^2 + 1 = 0$ in $\mathbb{R}$

Suppose that $x \in \mathbb{R}$. Since $\mathbb{R}$ is an ordered field, every square is nonnegative:

$$x^2 \geq 0.$$

Adding 1 and using $1 > 0$ gives

$$x^2 + 1 \geq 1 > 0.$$

In particular, $x^2 + 1$ cannot equal 0. Therefore

$$\boxed{x^2 + 1 = 0 \text{ has no solution in } \mathbb{R}.}$$

Completeness fills the gaps detected by Cauchy sequences, but it does not guarantee solutions to polynomial equations. We therefore need to enlarge $\mathbb{R}$ once more.

# Final picture

We proved that

- pre-reals are rational Cauchy sequences, modulo differences tending to zero;
- boundedness controls products, and a nonzero class stays uniformly away from zero, which controls reciprocals;
- the quotient $\mathbb{R}$ is a linearly ordered field containing an order-preserving copy of $\mathbb{Q}$;
- rational points are dense in $\mathbb{R}$;
- every real Cauchy sequence converges;
- the Newton Cauchy sequence defines a positive $\alpha \in \mathbb{R}$ satisfying $\alpha^2 = 2$;
- nevertheless, $x^2 + 1 = 0$ has no solution in $\mathbb{R}$.

# The complex numbers

# Why introduce the complex numbers?

We have just proved that the equation

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$. Equivalently, no real number has square $-1$, even though $\mathbb{R}$ is complete. We enlarge $\mathbb{R}$ by adjoining a symbol $i$ satisfying

$$\boxed{i^2 = -1.}$$

The resulting number system will

- be a field containing $\mathbb{R}$;
- carry an absolute value and be complete;
- contain a root of every nonconstant polynomial with complex coefficients.

# Definition of the complex numbers

## Definition

The set of complex numbers is

$$\boxed{\mathbb{C} = \mathbb{R}^2.}$$

We write the pair $(a, b)$ as $a + bi$, where

$$i = (0, 1).$$

Equality is componentwise:

$$a + bi = c + di \iff a = c \text{ and } b = d.$$

Thus every complex number has a unique *real part* and *imaginary part*,

$$\mathrm{Re}(a + bi) = a, \qquad \mathrm{Im}(a + bi) = b.$$

We identify $a \in \mathbb{R}$ with $a + 0i \in \mathbb{C}$.

# Addition and multiplication

For $z = a + bi$ and $w = c + di$, define

$$\begin{aligned} z + w &= (a + c) + (b + d)i, \\ zw &= (ac - bd) + (ad + bc)i. \end{aligned}$$

The multiplication formula is exactly what formal expansion and $i^2 = -1$ predict:

$$(a + bi)(c + di) = ac + (ad + bc)i + bd\, i^2.$$

The distinguished elements and additive inverse are

$$0 = 0 + 0i, \qquad 1 = 1 + 0i, \qquad -(a + bi) = (-a) + (-b)i.$$

In particular,

$$i^2 = (0, 1)(0, 1) = (-1, 0) = -1.$$

Restricted to the pairs $(a, 0)$, both operations reduce to real addition and multiplication, so the identification of $\mathbb{R}$ inside $\mathbb{C}$ preserves arithmetic.

# The commutative-ring laws

Addition is componentwise, so its associativity and commutativity follow immediately from the corresponding laws in $\mathbb{R}$.
Substituting the multiplication formula shows that

$$zw = wz, \qquad (zw)u = z(wu), \qquad z(w+u) = zw + zu, \qquad 1z = z.$$

Each real and imaginary component reduces to an identity in $\mathbb{R}$. For example, writing $u = e + fi$, both real components in the distributive law are

$$a(c+e) - b(d+f) = (ac - bd) + (ae - bf),$$

and the imaginary components agree in the same way.
Also $0 \neq 1$, because the pairs $(0,0)$ and $(1,0)$ are distinct. Hence these operations make $\mathbb{C}$ a nontrivial commutative ring.

# Conjugation and multiplicative inverses

Define the complex conjugate of $z = a + bi$ by

$$\overline{z} = a - bi.$$

Direct calculation gives

$$z\overline{z} = (a + bi)(a - bi) = a^2 + b^2.$$

If $z \neq 0$, then $a$ and $b$ are not both zero. Since squares are nonnegative in $\mathbb{R}$, this implies

$$a^2 + b^2 > 0.$$

We may therefore define

$$\boxed{z^{-1} = \frac{\overline{z}}{z\overline{z}} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.}$$

Then $zz^{-1} = 1$. Every nonzero element has an inverse, so

**Conclusion**

The complex numbers form a field containing $\mathbb{R}$.

# The complex field cannot be ordered

Suppose that a linear order made $\mathbb{C}$ into an ordered field. In any ordered field every square is nonnegative. Indeed, if $x \geq 0$, then $x^2 \geq 0$; if $x < 0$, then $-x > 0$ and $x^2 = (-x)^2 \geq 0$. Applied to $i$, this would give

$$-1 = i^2 \geq 0.$$

Adding 1 to both sides yields $0 \geq 1$. But $1 = 1^2 \geq 0$ and $1 \neq 0$, so $0 < 1$ in any ordered field — a contradiction. Therefore

$$\boxed{\text{no linear order makes } \mathbb{C} \text{ an ordered field.}}$$

This does not prevent us from measuring the *size* of a complex number; for that we use the real order.

# The complex absolute value

A standard consequence of the completeness of $\mathbb{R}$ is that every nonnegative real number $r$ has a unique nonnegative square root $\sqrt{r}$; the Newton iteration used for $x^2 = 2$ adapts to any $r \geq 0$.

**Definition**

For $z = a + bi \in \mathbb{C}$, define

$$\boxed{|z| = \sqrt{a^2 + b^2} \in \mathbb{R}.}$$

Equivalently,

$$|z|^2 = z\overline{z}, \qquad |z| \geq 0.$$

It follows at once that

$$|z| = 0 \iff z = 0, \qquad |\overline{z}| = |z|, \qquad |\operatorname{Re} z| \leq |z|, \qquad |\operatorname{Im} z| \leq |z|.$$

# Multiplicativity of the absolute value

Conjugation respects multiplication:

$$\overline{zw} = \overline{z}\,\overline{w}.$$

Hence

$$\begin{aligned} |zw|^2 &= (zw)\overline{(zw)} \\ &= z\overline{z}\,w\overline{w} \\ &= |z|^2|w|^2 = \big(|z|\,|w|\big)^2. \end{aligned}$$

Both $|zw|$ and $|z|\,|w|$ are nonnegative, so uniqueness of the nonnegative square root gives

$$\boxed{|zw| = |z|\,|w|.}$$

In particular, for $z \neq 0$,

$$|z^{-1}| = |z|^{-1}.$$

# The triangle inequality

For any complex number $u$, the coordinate formula gives

$$\operatorname{Re}(u) \le |u|.$$

Therefore

$$\begin{aligned} |z+w|^2 &= (z+w)(\overline{z}+\overline{w}) \\ &= |z|^2 + 2\operatorname{Re}(z\overline{w}) + |w|^2 \\ &\le |z|^2 + 2|z\overline{w}| + |w|^2 \\ &= |z|^2 + 2|z|\,|w| + |w|^2 = \big(|z| + |w|\big)^2, \end{aligned}$$

by multiplicativity: $|z\overline{w}| = |z|\,|\overline{w}| = |z|\,|w|$. Both sides before squaring are nonnegative, so

$$\boxed{|z+w| \le |z| + |w|.}$$

Thus

$$d(z,w) = |z-w|$$

is a metric on $\mathbb{C}$.

# Convergent and Cauchy sequences

Let $z = (z_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{C}$.

## Convergence

We say that $z$ converges to $w \in \mathbb{C}$ if

$$\forall \varepsilon > 0 \; \exists N \; \forall n \geq N, \qquad |z_n - w| \leq \varepsilon.$$

## Cauchy condition

We say that $z$ is Cauchy if

$$\forall \varepsilon > 0 \; \exists N \; \forall p, q \geq N, \qquad |z_p - z_q| \leq \varepsilon.$$

Here $\varepsilon$ is always a positive real number. The field $\mathbb{C}$ is called complete if every complex Cauchy sequence converges in $\mathbb{C}$.

# Every convergent sequence is Cauchy

Suppose $z_n \to z$, and let $\varepsilon > 0$. There is an $N$ such that

$$|z_n - z| \le \frac{\varepsilon}{2} \qquad (n \ge N).$$

If $p, q \ge N$, the triangle inequality gives

$$\begin{aligned} |z_p - z_q| &= |(z_p - z) + (z - z_q)| \\ &\le |z_p - z| + |z_q - z| \\ &\le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Hence $(z_n)$ is Cauchy.
The converse is precisely the completeness question. We now reduce it to the completeness of the two real coordinates.

# A complex Cauchy sequence has Cauchy coordinates

Let

$$z_n = a_n + b_n i$$

be Cauchy in $\mathbb{C}$. Since

$$|a_p - a_q| \leq |z_p - z_q|, \qquad |b_p - b_q| \leq |z_p - z_q|,$$

the real sequences $(a_n)$ and $(b_n)$ are Cauchy.
The completeness of $\mathbb{R}$ supplies real numbers $a, b$ such that

$$a_n \longrightarrow a, \qquad b_n \longrightarrow b.$$

The only possible complex limit is therefore

$$z = a + bi$$

(uniqueness of limits is left as an exercise). It remains to check convergence in the complex absolute value.

# The complex numbers are complete

Fix $\varepsilon > 0$. The two coordinate convergences give an $N$ such that, for $n \geq N$,

$$|a_n - a| \leq \frac{\varepsilon}{2}, \qquad |b_n - b| \leq \frac{\varepsilon}{2}.$$

For such $n$, the triangle inequality and $|ri| = |r|$ give

$$\begin{aligned} |z_n - (a + bi)| &= |(a_n - a) + (b_n - b)i| \\ &\leq |a_n - a| + |(b_n - b)i| \\ &= |a_n - a| + |b_n - b| \\ &\leq \varepsilon. \end{aligned}$$

Thus $z_n \to a + bi$.

**Completeness of $\mathbb{C}$**

Every complex Cauchy sequence converges: the completeness of $\mathbb{R}$, applied to the two coordinate sequences, proves the completeness of $\mathbb{C}$.

# The equations $x^2 + 1 = 0$ and $x^4 + 1 = 0$

By construction,

$$i^2 + 1 = 0,$$

so $x^2 + 1 = 0$ has the solutions $i$ and $-i$ in $\mathbb{C}$.
Set

$$\alpha = \frac{1+i}{\sqrt{2}}.$$

Since $(\sqrt{2})^2 = 2$,

$$\alpha^2 = \frac{(1+i)^2}{2} = \frac{1+2i+i^2}{2} = i, \qquad \alpha^4 = i^2 = -1.$$

Hence $\alpha^4 + 1 = 0$. In fact the four displayed choices

$$\boxed{\frac{1+i}{\sqrt{2}}, \quad \frac{-1-i}{\sqrt{2}}, \quad \frac{1-i}{\sqrt{2}}, \quad \frac{-1+i}{\sqrt{2}}}$$

all satisfy $x^4 + 1 = 0$.

# The Fundamental Theorem of Algebra

Let

$$P(z) = a_0 + a_1 z + \cdots + a_n z^n, \qquad a_n \neq 0, \quad n \geq 1,$$

be a nonconstant polynomial with complex coefficients. The Fundamental Theorem of Algebra states that there exists $\zeta \in \mathbb{C}$ such that

$$\boxed{P(\zeta) = 0.}$$

We state this theorem without proof. Despite its name it is not a purely algebraic fact: every proof uses the completeness of $\mathbb{R}$ in an essential way.

## Consequence

Dividing out one root at a time, every polynomial of degree $n$ factors completely:

$$P(z) = a_n(z - \zeta_1) \cdots (z - \zeta_n).$$

# Final picture

We proved that

- $\mathbb{C} = \mathbb{R}^2$, with multiplication determined by $i^2 = -1$, is a field containing $\mathbb{R}$;
- no linear order can make $\mathbb{C}$ an ordered field;
- $|a + bi| = \sqrt{a^2 + b^2}$ is multiplicative and satisfies the triangle inequality;
- convergence and the Cauchy condition are measured by this absolute value;
- the completeness of $\mathbb{R}$ in each coordinate proves that $\mathbb{C}$ is complete;
- $x^2 + 1 = 0$ and $x^4 + 1 = 0$ have explicit complex solutions.

We also stated, without proof, the Fundamental Theorem of Algebra: every nonconstant complex polynomial has a root in $\mathbb{C}$.

# The $p$-adic numbers

# How an infinite decimal expansion works

A real number is reached through its truncations. For example,

$$538, \quad 538.2, \quad 538.21, \quad 538.219, \quad 538.2194, \quad \ldots \quad \longrightarrow \quad 538.21947\ldots$$

Each step glues one more digit onto the *right*, changing the value by at most $9 \cdot 10^{-k}$ with $k$ growing. The truncations form a Cauchy sequence precisely because high *negative* powers of 10 are small: digits far to the right barely matter.

**The question behind this section**

That is a choice, not a necessity. What if we instead decide that digits far to the *left* barely matter?

# Growing to the left instead

Run the same process in the opposite direction:

$$3.706, \quad 53.706, \quad 153.706, \quad 4153.706, \quad 84153.706, \quad \ldots \quad \longrightarrow \quad \ldots 84153.706$$

Each step glues one more digit onto the *left*. Writing $a_j$ for the digit multiplying $10^j$, the step from the truncation $x_k$ to a later truncation $x_\ell$ adds only digits in positions $k \le j < \ell$:

$$x_\ell - x_k = \sum_{j=k}^{\ell-1} a_j\, 10^j = 10^k \cdot (\text{an integer}).$$

## The guiding idea

In a distance where high *positive* powers of 10 are small, these truncations also form a Cauchy sequence, and the left-infinite string denotes its limit.

# Real versus $n$-adic expansions

The same idea works in any base $n \geq 2$, with digits $0, 1, \ldots, n-1$.

## Real expansions

A real base-$n$ expansion has only finitely many digits *left* of the point, but may continue forever to the right:

$$\pm a_m \cdots a_1 a_0 . a_{-1} a_{-2} a_{-3} \cdots$$

## $n$-adic expansions

An $n$-adic expansion may continue forever to the *left*, but has only finitely many digits to the right:

$$\cdots a_3 a_2 a_1 a_0 . a_{-1} \cdots a_{-s}$$

Each system allows infinitely many digits on exactly one side of the point. In base 10, compare

$$\pm 538.21947\ldots \qquad \text{with} \qquad \ldots 84153.706.$$

Note the missing $\pm$ on the right: an $n$-adic expansion carries no sign. We will see shortly why none is needed.

# Arithmetic still runs from right to left

In a left-infinite decimal expansion, addition is performed by the usual digit algorithm:

$$\begin{array}{r} \ldots 84153.706 \\ + \quad \ldots 70926.153 \\ \hline \ldots 55079.859 \end{array}$$

Each requested final block of digits depends on only finitely many positions to its right. Equivalently, for every $k$, ordinary arithmetic determines the answer modulo $10^k$. Multiplication works in the same way: compute to any chosen finite precision, and increasing the precision does not change the digits already determined near the point.

## Digit strings behave better than real expansions

Two left-infinite strings are equal exactly when *all* their digits agree; there is no analogue of $0.999\ldots = 1.000\ldots$. Moreover, each digit of a sum or product depends on finitely many digits of the inputs. For real expansions this fails: in $0.333\ldots + 0.666\ldots$, no finite number of digits decides whether the integer part is 0 or 1.

# Negative numbers acquire infinite left tails

The finite decimal strings

$$9, \qquad 99, \qquad 999, \qquad 9999, \quad \dots$$

approach $-1$ in the 10-adic sense. Indeed,

$$(10^k - 1) - (-1) = 10^k,$$

and a high power of 10 will have very small 10-adic size. Thus

$$\boxed{-1 = \dots 999999.}$$

The digit algorithm agrees: adding 1 to $\dots 999999$ launches a carry that travels forever to the left, so $\dots 999999 + 1 = \dots 000000 = 0$.
Similarly,

$$83, \qquad 983, \qquad 9983, \qquad 99983, \quad \dots$$

approach $-17$, because $10^k - 17$ differs from $-17$ by $10^k$. Hence

$$\boxed{-17 = \dots 999983.}$$

This explains the missing sign: a left-infinite string needs no $\pm$, because negatives are already ordinary digit strings — the infinite tail of 9s does the work of the minus sign.

# A rational left-infinite expansion

Solving $7x = 3$ modulo successive powers of 10 gives

$$9, \quad 29, \quad 429, \quad 1429, \quad 71429, \quad 571429, \quad 8571429, \ldots$$

For instance, $7 \cdot 1429 = 10003$ and $7 \cdot 8571429 = 60000003$: the products agree with 3 in the final four and seven digits.
These compatible solutions converge 10-adically to

$$\boxed{\frac{3}{7} = \ldots 8571428571429.}$$

The digit algorithm confirms this: multiplying $\ldots 8571429$ by 7 right to left starts with $7 \cdot 9 = 63$, and every later step produces a multiple of 10, so the product is $\ldots 0000003 = 3$.
Dividing by 10 merely shifts the point:

$$\frac{3}{70} = \ldots 857142857142.9.$$

Rational numbers therefore appear as eventually periodic left-infinite expansions, just as they appear as eventually periodic real expansions to the right.

# Which digits control distance?

In the ordinary metric, agreement far to the *right* of the point means closeness. For example,

$$|8.21947\ldots - 8.21929\ldots| = 0.00018\ldots < 10^{-3}.$$

In the 10-adic metric, closeness means agreement in the fractional digits and in many final digits *left* of the point:

$$\begin{aligned} x &= \ldots 84153.7, \\ y &= \ldots 26153.7. \end{aligned} \qquad x - y = \ldots 58000.0.$$

Their difference is divisible by $10^3$ and by no higher power of 10, so their 10-adic distance will be

$$\|x - y\|_{10} = 10^{-3}.$$

This is the bridge from digit strings to the formal definition: count the powers of the base dividing a difference.

# From base 10 to a prime base

The opening slides used base 10 because its digits are familiar. For the formal theory we now fix a prime $p$ and work in base $p$.

## Why a prime base?

Everything below can be generalized to an arbitrary base $n \geq 2$, but the prime case is simpler: divisibility is governed by a single prime, and the resulting size is *multiplicative*.

From the arithmetic section: $\nu_p(m)$ counts the factors $p$ in the unique prime factorization of $m$, and

$$\nu_p(m_1 m_2) = \nu_p(m_1) + \nu_p(m_2).$$

We set $\nu_p(-m) = \nu_p(m)$, so that $\nu_p$ is defined on all nonzero integers.

# The $p$-adic order

## Definition

For $x \in \mathbb{Q}^\times$, write $x = a/b$ with nonzero integers $a, b$ and define the $p$-adic order

$$\boxed{\nu_p(x) = \nu_p(a) - \nu_p(b), \qquad \nu_p(0) = +\infty.}$$

This is well defined: if $a/b = c/d$, then $ad = bc$, and additivity gives

$$\nu_p(a) + \nu_p(d) = \nu_p(b) + \nu_p(c),$$

so the two differences agree.
Additivity persists on $\mathbb{Q}^\times$:

$$\nu_p(xy) = \nu_p(x) + \nu_p(y).$$

# Order measures common final digits

For a nonzero integer $a$ and $k \in \mathbb{N}$, directly from unique factorization,

$$\boxed{p^k \mid a \quad \Longleftrightarrow \quad \nu_p(a) \geq k.}$$

In particular,

$$\nu_p(p^k) = k,$$

and $\nu_p(a)$ is exactly the largest exponent $k$ for which $p^k \mid a$.
If two integers have the same final $k$ base-$p$ digits, their difference is divisible by $p^k$, and hence has $p$-adic order at least $k$.

# The $p$-adic size

## Definition

For $x \in \mathbb{Q}$, define

$$\boxed{\|x\|_p = \begin{cases} 0, & x = 0, \\ p^{-\nu_p(x)}, & x \neq 0. \end{cases}}$$

The digit interpretation becomes

$$p^k \mid a \quad \Longleftrightarrow \quad \|a\|_p \leq p^{-k} \qquad (a \in \mathbb{Z}).$$

For example, with $p = 5$,

$$\|58000\|_5 = 5^{-3}, \qquad \|5^k\|_5 = 5^{-k}, \qquad \|58001\|_5 = 1.$$

This confirms the intuition: adding base-$p$ digits farther to the left changes a truncation by a $p$-adically smaller amount.

# The order laws

If $p^m$ divides two integers, it divides their sum. Writing $x = a/c$ and $y = b/c$ over a common denominator, this yields

$$\boxed{\nu_p(x+y) \geq \min\{\nu_p(x), \nu_p(y)\}.}$$

Indeed $\nu_p(a+b) \geq \min\{\nu_p(a), \nu_p(b)\}$; subtract $\nu_p(c)$ from both sides.
Multiplication is even better: additivity is an *equality*,

$$\boxed{\nu_p(xy) = \nu_p(x) + \nu_p(y).}$$

# Properties of the $p$-adic size

Translating the order laws gives

$$\|x\|_p \geq 0, \qquad \|x\|_p = 0 \iff x = 0, \qquad \|-x\|_p = \|x\|_p,$$

together with the ultrametric inequality

$$\boxed{\|x+y\|_p \leq \max\{\|x\|_p, \|y\|_p\}}$$

and multiplicativity

$$\boxed{\|xy\|_p = \|x\|_p \, \|y\|_p.}$$

The usual triangle and reverse-triangle estimates follow:

$$\|x+y\|_p \leq \|x\|_p + \|y\|_p, \qquad \big|\|x\|_p - \|y\|_p\big| \leq \|x-y\|_p.$$

In particular, $\|x^{-1}\|_p = \|x\|_p^{-1}$ for $x \neq 0$.

# The dominant term and the metric

The ultrametric inequality has a useful strengthening:

$$\|x\|_p \neq \|y\|_p \quad \Longrightarrow \quad \boxed{\|x+y\|_p = \max\{\|x\|_p, \|y\|_p\}.}$$

Indeed, suppose $\nu_p(x) < \nu_p(y)$. The sum law gives $\nu_p(x+y) \geq \nu_p(x)$; applied to $x = (x+y) - y$, it gives $\nu_p(x) \geq \min\{\nu_p(x+y), \nu_p(y)\}$, and since $\nu_p(y) > \nu_p(x)$ this forces $\nu_p(x+y) = \nu_p(x)$.
Define

$$d_p(x,y) = \|x-y\|_p.$$

Then $d_p$ is an ultrametric:

$$d_p(x,z) \leq \max\{d_p(x,y), d_p(y,z)\}.$$

We now complete $\mathbb{Q}$ with respect to this distance.

# $p$-adic Cauchy sequences

Let $x = (x_k)_{k\in\mathbb{N}}$ be a sequence of rational numbers.

**Definition**

The sequence $x$ is *$p$-adically Cauchy* if

$$\forall \varepsilon > 0\ \exists N\ \forall k, \ell \geq N, \qquad \|x_k - x_\ell\|_p \leq \varepsilon.$$

Write

$$\mathcal{C}_p = \{x \colon \mathbb{N} \to \mathbb{Q} : x \text{ is } p\text{-adically Cauchy}\}.$$

The truncations of a left-infinite base-$p$ string lie in $\mathcal{C}_p$: if $\ell > k$, their difference is divisible by $p^k$, so

$$\|x_\ell - x_k\|_p \leq p^{-k} \longrightarrow 0.$$

# Cauchy sequences are $p$-adically bounded

Let $x \in \mathcal{C}_p$. Choose $N \geq 1$ such that

$$\|x_k - x_\ell\|_p \leq 1 \qquad (k, \ell \geq N).$$

For $k \geq N$, the ultrametric inequality gives

$$\|x_k\|_p \leq \max\{\|x_k - x_N\|_p, \|x_N\|_p\} \leq \max\{1, \|x_N\|_p\}.$$

Taking the maximum of this bound and the finitely many initial sizes gives $B > 0$ such that

$$\boxed{\|x_k\|_p \leq B \qquad \text{for every } k.}$$

This is the same boundedness argument used for real Cauchy sequences, with the ultrametric maximum replacing a sum.

# Equivalent Cauchy sequences

## Definition

For $x, y \in \mathcal{C}_p$, define
$$x \sim_p y \quad \Longleftrightarrow \quad \|x_k - y_k\|_p \longrightarrow 0.$$

Reflexivity and symmetry are immediate. If $x \sim_p y$ and $y \sim_p z$, then

$$\|x_k - z_k\|_p \leq \max\{\|x_k - y_k\|_p, \|y_k - z_k\|_p\} \longrightarrow 0.$$

Thus $\sim_p$ is an equivalence relation.
Two digit strings represent the same class precisely when their truncations eventually agree to every fixed finite precision.

# Definition of the $p$-adic numbers

## Definition

The set of *$p$-adic numbers* is

$$\boxed{\mathbb{Q}_p = \mathcal{C}_p/\!\sim_p.}$$

We write $X = [x]$ when $X$ is represented by the $p$-adic Cauchy sequence $x$.

Define constants and arithmetic pointwise:

$$(-x)_k = -x_k, \qquad (x+y)_k = x_k + y_k, \qquad (xy)_k = x_k y_k.$$

Therefore

$$-[x] = [-x], \qquad [x]+[y] = [x+y], \qquad [x][y] = [xy].$$

# Why arithmetic descends to the quotient

Negation and addition preserve Cauchy sequences directly. For products, boundedness and multiplicativity give

$$\|x_k y_k - x_\ell y_\ell\|_p \le \max\{\|x_k\|_p \, \|y_k - y_\ell\|_p, \\ \|y_\ell\|_p \, \|x_k - x_\ell\|_p\}.$$

The right side tends to zero as $k, \ell \to \infty$.
The same estimate proves that equivalent representatives give equivalent sums and products. All commutative-ring laws hold pointwise before passing to classes.

**Conclusion**

For every prime $p$, $\mathbb{Q}_p$ is a commutative ring.

# The rational numbers inside $\mathbb{Q}_p$

A rational number determines a constant Cauchy sequence. Define

$$\iota_p \colon \mathbb{Q} \longrightarrow \mathbb{Q}_p, \qquad \iota_p(r) = [(r, r, r, \ldots)].$$

Pointwise arithmetic makes $\iota_p$ a ring homomorphism.
If $\iota_p(r) = \iota_p(s)$, then the constant sequence $\|r - s\|_p$ tends to zero. If $r \neq s$, choosing

$$\varepsilon = \frac{\|r - s\|_p}{2} > 0$$

makes this impossible. Therefore $r = s$, and $\iota_p$ is injective.
We may consequently regard $\mathbb{Q}$ as a subring of $\mathbb{Q}_p$. In particular, $\mathbb{Q}_p$ is nontrivial.

# Reciprocals in $\mathbb{Q}_p$

Let $X = [x] \neq 0$. Since $x \not\sim_p 0$, there is $\delta > 0$ with $\|x_k\|_p > \delta$ for infinitely many $k$; choose $N$ such that $\|x_k - x_\ell\|_p \leq \delta$ for $k, \ell \geq N$. The dominant-term lemma then freezes the size along the tail: there is $c > \delta$ with

$$\|x_k\|_p = c \qquad (k \geq N).$$

In particular $x_k \neq 0$ there, and the tail $y_k = x_k^{-1}$ is Cauchy, because multiplicativity gives

$$\|y_k - y_\ell\|_p = \frac{\|x_k - x_\ell\|_p}{\|x_k\|_p \, \|x_\ell\|_p} = \frac{\|x_k - x_\ell\|_p}{c^2}.$$

The finitely many earlier values are irrelevant; since $x_k y_k = 1$ on the tail, the class $Y = [y]$ satisfies $XY = 1$.

## Conclusion

Every nonzero $p$-adic number is invertible: $\mathbb{Q}_p$ is a field containing $\mathbb{Q}$.

# Extending the $p$-adic size to $\mathbb{Q}_p$

If $x \in \mathcal{C}_p$, the reverse-triangle estimate gives

$$\big| \|x_k\|_p - \|x_\ell\|_p \big| \le \|x_k - x_\ell\|_p.$$

Thus $(\|x_k\|_p)$ is a Cauchy sequence of real numbers and has a real limit.

## Definition

For $X = [x] \in \mathbb{Q}_p$, define

$$\boxed{\|X\|_p = \lim_{k\to\infty} \|x_k\|_p.}$$

If $x \sim_p y$, the same estimate shows that the two real limits agree. Hence the definition is independent of the representative.

Passing the rational identities and inequalities to limits gives

$$\|X\|_p \geq 0, \qquad \|X\|_p = 0 \iff X = 0, \qquad \|-X\|_p = \|X\|_p,$$

$$\|XY\|_p = \|X\|_p \, \|Y\|_p, \qquad \|X+Y\|_p \leq \max\{\|X\|_p, \|Y\|_p\}.$$

In particular, multiplicativity survives the limit.
Thus

$$d_p(X, Y) = \|X - Y\|_p$$

is an ultrametric on $\mathbb{Q}_p$. The rational embedding preserves distance:

$$\|\iota_p(r) - \iota_p(s)\|_p = \|r - s\|_p.$$

# The rational numbers are dense in $\mathbb{Q}_p$

Let $X = [x] \in \mathbb{Q}_p$ and fix $\varepsilon > 0$. Since $x$ is Cauchy, choose $N$ such that

$$\|x_k - x_N\|_p \leq \frac{\varepsilon}{2} \qquad (k \geq N).$$

The difference $X - \iota_p(x_N)$ is represented by $(x_k - x_N)_k$, so

$$\|X - \iota_p(x_N)\|_p = \lim_{k\to\infty} \|x_k - x_N\|_p \leq \frac{\varepsilon}{2} < \varepsilon.$$

Therefore

$$\boxed{\forall X \in \mathbb{Q}_p \ \forall \varepsilon > 0 \ \exists q \in \mathbb{Q}, \qquad \|X - \iota_p(q)\|_p < \varepsilon.}$$

# Completeness: rational approximations

Let $(X_k)_{k\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{Q}_p$, Cauchy and convergent sequences in $\mathbb{Q}_p$ being defined as in $\mathbb{C}$, with $\|\cdot\|_p$ in place of the absolute value. By rational density, choose $q_k \in \mathbb{Q}$ such that

$$\|X_k - \iota_p(q_k)\|_p < p^{-k}.$$

The rational sequence $q = (q_k)$ is $p$-adically Cauchy, because

$$\|q_k - q_\ell\|_p \leq \max\{\|\iota_p(q_k) - X_k\|_p, \|X_k - X_\ell\|_p, \\ \|X_\ell - \iota_p(q_\ell)\|_p\},$$

and all three terms tend to zero as $k, \ell \to \infty$.
Hence $q \in \mathcal{C}_p$, and it defines

$$X = [q] \in \mathbb{Q}_p.$$

# Every Cauchy sequence converges

Because $q = (q_k)$ is Cauchy,

$$\iota_p(q_k) \longrightarrow [q] = X.$$

Indeed, for every $\varepsilon > 0$, all sufficiently large $k, \ell$ satisfy

$$\|q_\ell - q_k\|_p \leq \frac{\varepsilon}{2},$$

and therefore

$$\|X - \iota_p(q_k)\|_p = \lim_{\ell \to \infty} \|q_\ell - q_k\|_p \leq \frac{\varepsilon}{2} < \varepsilon.$$

Finally,

$$\|X_k - X\|_p \leq \max\{\|X_k - \iota_p(q_k)\|_p, \|\iota_p(q_k) - X\|_p\} \longrightarrow 0.$$

**Completeness of $\mathbb{Q}_p$**

Every $p$-adic Cauchy sequence in $\mathbb{Q}_p$ converges in $\mathbb{Q}_p$.

# The closed unit ball

The field $\mathbb{Q}_p$ contains elements of arbitrarily large and small $p$-adic size. We single out those of size at most 1.

## Definition

The *$p$-adic integers* form the closed unit ball

$$\boxed{\mathbb{Z}_p = \{X \in \mathbb{Q}_p : \|X\|_p \leq 1\} = \overline{B}(0,1).}$$

For a rational number $x$, this condition has the familiar description

$$x \in \mathbb{Q} \cap \mathbb{Z}_p \quad \Longleftrightarrow \quad \nu_p(x) \geq 0.$$

Informally, these are the $p$-adic numbers whose base-$p$ expansion has no digits to the right of the point — a picture made precise at the end of this section.

# Why the unit ball is a ring

Certainly $0, 1 \in \mathbb{Z}_p$. If $X, Y \in \mathbb{Z}_p$, the ultrametric and multiplicative laws give

$$\|X - Y\|_p \leq \max\{\|X\|_p, \|Y\|_p\} \leq 1, \qquad \|XY\|_p = \|X\|_p \|Y\|_p \leq 1.$$

Thus $X - Y$ and $XY$ again lie in $\mathbb{Z}_p$.

**Conclusion**

The unit ball $\mathbb{Z}_p$ is a subring of the field $\mathbb{Q}_p$.

This is a genuinely $p$-adic phenomenon. The real unit ball is

$$\{x \in \mathbb{R} : |x| \leq 1\} = [-1, 1],$$

which is not a ring: it contains 1, but not $1 + 1 = 2$. The ordinary triangle inequality only bounds a sum by 2; the ultrametric inequality keeps it inside the same unit ball.

# The ordinary integers inside $\mathbb{Z}_p$

If $n \in \mathbb{Z}$ is nonzero, then $\nu_p(n)$ counts the copies of $p$ in its prime factorization. In particular,

$$\nu_p(n) \geq 0 \qquad \Longrightarrow \qquad \|n\|_p = p^{-\nu_p(n)} \leq 1.$$

The same is immediate for $n = 0$. Hence the embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ restricts to an inclusion

$$\boxed{\mathbb{Z} \subseteq \mathbb{Z}_p.}$$

There are two useful possibilities:

$$\|n\|_p = \begin{cases} 1, & p \nmid n, \\ p^{-k}, & p^k \mid n \text{ but } p^{k+1} \nmid n. \end{cases}$$

Thus divisibility by higher powers of $p$ makes an ordinary integer progressively smaller inside $\mathbb{Z}_p$.

# The rational numbers inside $\mathbb{Z}_p$

## Definition

Let $\mathbb{Z}_{(p)}$ be the set of fractions whose denominator is not divisible by $p$:

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} : a, b \in \mathbb{Z},\ p \nmid b \right\}.$$

A denominator not divisible by $p$ has $\nu_p(b) = 0$. For a nonzero fraction of this form,

$$\left\| \frac{a}{b} \right\|_p = p^{-(\nu_p(a) - \nu_p(b))} = p^{-\nu_p(a)} \leq 1,$$

and so

$$\boxed{\mathbb{Z} \subseteq \mathbb{Z}_{(p)} \subseteq \mathbb{Z}_p.}$$

In fact these are exactly the rational points of the unit ball:

$$\boxed{\mathbb{Q} \cap \mathbb{Z}_p = \mathbb{Z}_{(p)}.}$$

Indeed, after a fraction is reduced, a $p$ in its denominator would give it negative $p$-adic order.

# The $p$-adic integers are complete

The size map $X \mapsto \|X\|_p$ is continuous: the reverse-triangle estimate, valid in $\mathbb{Q}_p$ exactly as in $\mathbb{Q}$, gives

$$\big|\|X\|_p - \|Y\|_p\big| \leq \|X - Y\|_p.$$

Consequently,

$$\mathbb{Z}_p = \{X \in \mathbb{Q}_p : \|X\|_p \leq 1\}$$

is a closed subspace of $\mathbb{Q}_p$.
We already know that $\mathbb{Q}_p$ is complete, and every closed subspace of a complete metric space is complete. Therefore

$$\boxed{\mathbb{Z}_p \text{ is complete.}}$$

Concretely, a Cauchy sequence of $p$-adic integers has a limit in $\mathbb{Q}_p$; closedness says that this limit still has size at most 1, and hence still belongs to $\mathbb{Z}_p$.

# The ordinary integers are dense in $\mathbb{Z}_p$

Let $X \in \mathbb{Z}_p$ and $N \in \mathbb{N}$. Rational density in $\mathbb{Q}_p$ gives $q \in \mathbb{Q}$ with $\|X - q\|_p \leq p^{-N}$, and the ultrametric inequality keeps the approximation inside the unit ball:

$$\|q\|_p \leq \max\{\|X\|_p, \|q - X\|_p\} \leq 1.$$

Hence $q \in \mathbb{Q} \cap \mathbb{Z}_p = \mathbb{Z}_{(p)}$: write $q = a/b$ with $p \nmid b$.
Since $b$ and $p^N$ have no common prime factor, Bézout's identity provides $u, v \in \mathbb{Z}$ with $ub + vp^N = 1$.
The integer $c = ua$ then satisfies

$$q - c = \frac{a(1 - ub)}{b} = \frac{av}{b}\, p^N, \qquad \text{so} \qquad \|q - c\|_p \leq p^{-N},$$

and one more ultrametric step gives $\|X - c\|_p \leq p^{-N}$.

## A non-Archimedean surprise

The ordinary integers are dense in $\mathbb{Z}_p$. Compare with $\mathbb{R}$, where $\mathbb{Z}$ is discrete and closed — as far from dense as a set can be.

# The canonical remainder at each level

Fix $X \in \mathbb{Z}_p$ and $N \geq 1$. Among all integer approximations of $X$ there is a preferred one: exactly one

$$r_N \in \{0, 1, \ldots, p^N - 1\} \qquad \text{satisfies} \qquad \|X - r_N\|_p \leq p^{-N}.$$

*Existence.* Density gives $c \in \mathbb{Z}$ with $\|X - c\|_p \leq p^{-N}$; let $r_N$ be the remainder of $c$ upon division by $p^N$. The discarded multiple of $p^N$ has size at most $p^{-N}$, so the ultrametric inequality keeps $r_N$ within $p^{-N}$ of $X$.

*Uniqueness.* Two candidates differ by an ordinary integer of size at most $p^{-N}$, that is, by a multiple of $p^N$; both lying in $\{0, \ldots, p^N - 1\}$, they are equal.

Uniqueness also forces compatibility: the remainder of $r_{N+1}$ modulo $p^N$ approximates $X$ within $p^{-N}$ as well, whence

$$\boxed{r_{N+1} \equiv r_N \pmod{p^N}.}$$

# From remainders to digits

Compatibility means $r_{N+1} = r_N + a_N p^N$ for a unique digit $a_N \in \{0, 1, \ldots, p-1\}$. The canonical remainders therefore unfold digit by digit,

$$r_N = a_0 + a_1 p + a_2 p^2 + \cdots + a_{N-1} p^{N-1},$$

and $\|X - r_N\|_p \leq p^{-N} \longrightarrow 0$ says exactly that

$$\boxed{X = a_0 + a_1 p + a_2 p^2 + a_3 p^3 + \cdots}$$

as a limit of truncations.

**The opening picture, now justified**

Every $p$-adic integer is a left-infinite base-$p$ digit string $\ldots a_3 a_2 a_1 a_0$, and distinct strings represent distinct $p$-adic integers, by the uniqueness of the canonical remainders.

# A purely arithmetic description of $\mathbb{Z}_p$

Collect the canonical remainders of a $p$-adic integer into a single sequence. This yields a bijection, which we state without proof:

$$\boxed{\mathbb{Z}_p \longleftrightarrow \left\{(r_1, r_2, r_3, \ldots) : 0 \leq r_n < p^n \text{ and } r_{n+1} \equiv r_n \pmod{p^n}\right\},}$$

under which sums and products are computed level by level: add or multiply the $n$-th remainders, then reduce modulo $p^n$.

Why the two sides carry the same information:

- two $p$-adic integers with the same remainders at every level lie within $p^{-n}$ of each other for all $n$, hence coincide;
- any compatible sequence satisfies $r_m \equiv r_n \pmod{p^n}$ for $m \geq n$, so it is $p$-adically Cauchy, and its limit is a $p$-adic integer whose canonical remainders are the given ones;
- level-by-level arithmetic is the digit algorithm of the opening slides: any finite precision of a sum or product needs only finitely many digits of the inputs.

The right-hand side never mentions distances or Cauchy sequences: a $p$-adic integer *is* a coherent choice of remainder modulo every power of $p$.

# A polynomial without roots in $\mathbb{Q}_p$

While constructing reciprocals we saw that for a nonzero $X = [x]$ the sizes $\|x_k\|_p$ are eventually constant. The limit defining $\|X\|_p$ is therefore one of these rational sizes:

$$\boxed{\|X\|_p = p^{-m} \text{ for some } m \in \mathbb{Z} \qquad (X \neq 0).}$$

The completion created no new sizes.
Now suppose $Y \in \mathbb{Q}_p$ satisfied $Y^2 = p$. Multiplicativity would give

$$\|Y\|_p^2 = \|p\|_p = p^{-1}, \qquad \text{so} \qquad \|Y\|_p = p^{-1/2},$$

which is not an integer power of $p$ — impossible.

**$\mathbb{Q}_p$ is not algebraically closed**

The polynomial $x^2 - p$ has no root in $\mathbb{Q}_p$.

In $\mathbb{R}$, the rootless polynomial $x^2 + 1$ was detected by the *order*: squares are never negative. In $\mathbb{Q}_p$, the rootless polynomial $x^2 - p$ is detected by the *size*.

# Adjoining a square root is not enough

Nothing forbids enlarging $\mathbb{Q}_p$ exactly as we enlarged $\mathbb{R}$:

$$\mathbb{Q}_p(\sqrt{p}) = \{A + B\sqrt{p} : A, B \in \mathbb{Q}_p\},$$

with multiplication dictated by $(\sqrt{p})^2 = p$, just as $i^2 = -1$ dictated multiplication in $\mathbb{C}$. The conjugate trick $(A + B\sqrt{p})(A - B\sqrt{p}) = A^2 - pB^2$ produces reciprocals — the denominator is nonzero precisely because $p$ has no square root — and the size extends, with $\|\sqrt{p}\|_p = p^{-1/2}$. Sizes in $\mathbb{Q}_p(\sqrt{p})$ are now the powers of $p^{1/2}$.

In the real case this single step was miraculous: adjoining one root of one polynomial produced $\mathbb{C}$, where *every* nonconstant polynomial has a root.

## The miracle does not repeat

The size argument simply moves one level up: a root of $x^4 - p$ would have size $p^{-1/4}$, which is not a power of $p^{1/2}$. So $x^4 - p$ has no root in $\mathbb{Q}_p(\sqrt{p})$.

Adjoining a fourth root of $p$ repairs this and breaks again at $x^8 - p$: no finite number of adjunctions can finish the job.

# A gigantic extension

Repairing every failure at once requires a leap. One can construct a field $\overline{\mathbb{Q}_p} \supseteq \mathbb{Q}_p$, gathered from the roots of *all* polynomials, such that

- every nonconstant polynomial with coefficients in $\overline{\mathbb{Q}_p}$ has a root in $\overline{\mathbb{Q}_p}$ — the field is algebraically closed;
- the $p$-adic size extends to $\overline{\mathbb{Q}_p}$, its nonzero values now being *all* powers $p^r$ with $r$ rational.

We state its existence without proof.

**The price of the leap**

Unlike the two-dimensional step from $\mathbb{R}$ to $\mathbb{C}$, this extension is infinite-dimensional over $\mathbb{Q}_p$, and it is not complete: it contains Cauchy sequences with no limit.

Algebra has been repaired, but analysis is broken again.

# Completing once more: the field $\mathbb{C}_p$

We know exactly what to do with an incomplete field carrying a size: run the Cauchy construction a third time, now on $\overline{\mathbb{Q}_p}$.

### Definition

The field of *p-adic complex numbers* is the completion

$$\boxed{\mathbb{C}_p = (\text{Cauchy sequences in } \overline{\mathbb{Q}_p})/(\text{differences tending to } 0).}$$

One might fear an endless alternation — complete, close algebraically, complete again, ... A remarkable theorem, which we state without proof, says that the process stops here.

### Theorem (without proof)

$\mathbb{C}_p$ is complete *and* algebraically closed.

Thus $\mathbb{C}_p$ stands to $\mathbb{Q}_p$ as $\mathbb{C}$ stands to $\mathbb{R}$: a complete, algebraically closed field.

# The $p$-adic picture

For a fixed prime $p$, we proved that

- left-infinite digit strings are understood through compatible finite truncations;
- $\nu_p$ counts factors of $p$, and $\|x\|_p = p^{-\nu_p(x)}$ measures $p$-adic size;
- this size is ultrametric and multiplicative;
- $\mathbb{Q}_p$ is the quotient of rational $p$-adic Cauchy sequences by differences tending to zero;
- pointwise arithmetic makes $\mathbb{Q}_p$ a complete field containing a dense copy of $\mathbb{Q}$;
- its closed unit ball $\mathbb{Z}_p$ is a complete subring containing $\mathbb{Z}_{(p)}$, and $\mathbb{Z}$ is dense in $\mathbb{Z}_p$;
- a $p$-adic integer amounts to a compatible choice of remainders modulo every power of $p$ — equivalently, a left-infinite base-$p$ digit string;
- $x^2 - p$ has no root in $\mathbb{Q}_p$: completeness does not make the field algebraically closed.

We also glimpsed, without proof, the remedy: adjoining the roots of all polynomials and completing once more produces $\mathbb{C}_p$, a field both complete and algebraically closed — the $p$-adic counterpart of $\mathbb{C}$.

# Arithmetic modulo $n$

# Congruence and residue classes

Fix an integer $n \geq 2$. Two integers are congruent modulo $n$ when they have the same remainder after division by $n$:

$$a \equiv b \pmod{n} \qquad \Longleftrightarrow \qquad n \mid (a-b).$$

This is an equivalence relation. The equivalence class of $a$ is

$$\overline{a} = a + n\mathbb{Z} = \{a + kn : k \in \mathbb{Z}\}.$$

## Definition

The integers modulo $n$ form the set of residue classes

$$\boxed{\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}.}$$

Thus $\overline{a} = \overline{b}$ precisely when $a \equiv b \pmod{n}$. Every class has infinitely many integer representatives, but exactly one in $\{0, \dots, n-1\}$.

# Arithmetic modulo $n$

Define addition and multiplication using representatives:

$$\overline{a} + \overline{b} = \overline{a+b}, \qquad \overline{a}\,\overline{b} = \overline{ab}.$$

These operations are well defined. Indeed, if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$(a+b) - (a'+b') = (a-a') + (b-b')$$

and

$$ab - a'b' = a(b-b') + b'(a-a')$$

are both divisible by $n$.

## The quotient ring

With these operations, $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring:

$$0 = \overline{0}, \qquad 1 = \overline{1}, \qquad -\overline{a} = \overline{-a}.$$

Its ring laws are inherited from $\mathbb{Z}$.

The map $\pi_n \colon \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $a \mapsto \overline{a}$, preserves addition, multiplication, 0, and 1, and has kernel $n\mathbb{Z}$.

# Which residue classes are invertible?

A class $\overline{a}$ is a unit if some $\overline{b}$ satisfies $\overline{a}\,\overline{b} = \overline{1}$.

## Unit criterion

$$\boxed{\overline{a} \text{ is a unit in } \mathbb{Z}/n\mathbb{Z} \iff \gcd(a, n) = 1.}$$

If $\gcd(a, n) = 1$, Bézout's identity gives $u, v \in \mathbb{Z}$ with

$$au + nv = 1.$$

Reducing modulo $n$ yields $\overline{a}\,\overline{u} = \overline{1}$, so $\overline{u} = \overline{a}^{-1}$.
Conversely, if $\overline{a}\,\overline{b} = \overline{1}$, then $ab + kn = 1$ for some $k \in \mathbb{Z}$, and hence $\gcd(a, n) = 1$.
For example, in $\mathbb{Z}/10\mathbb{Z}$ the units are

$$\overline{1},\ \overline{3},\ \overline{7},\ \overline{9}; \qquad \overline{3}^{-1} = \overline{7}.$$

# Zero divisors

A nonzero element $x$ of a commutative ring is a zero divisor if $xy = 0$ for some nonzero $y$.

## Zero-divisor criterion

For $\overline{a} \neq \overline{0}$ in $\mathbb{Z}/n\mathbb{Z}$,

$$\boxed{\overline{a} \text{ is a zero divisor} \quad \Longleftrightarrow \quad \gcd(a, n) > 1.}$$

If $d = \gcd(a, n) > 1$, then $\overline{n/d} \neq \overline{0}$ and

$$\overline{a}\,\overline{n/d} = \overline{(a/d)n} = \overline{0}.$$

Conversely, if $\gcd(a, n) = 1$ then $\overline{a}$ is a unit, and no unit is a zero divisor: multiplying $\overline{a}\,\overline{y} = \overline{0}$ by $\overline{a}^{-1}$ forces $\overline{y} = \overline{0}$. Hence every class is exactly one of $\overline{0}$, a unit, or a zero divisor — the trichotomy governed by $\gcd(a, n)$.

Consequently,

$$\boxed{\mathbb{Z}/n\mathbb{Z} \text{ has a nonzero zero divisor} \quad \Longleftrightarrow \quad n \text{ is composite.}}$$

For instance, $\overline{2}\,\overline{3} = \overline{0}$ in $\mathbb{Z}/6\mathbb{Z}$, although neither factor is zero.

# When is $\mathbb{Z}/n\mathbb{Z}$ a field?

## The prime-modulus theorem

For $n \geq 2$, the following are equivalent:

$$\boxed{\begin{array}{c} \mathbb{Z}/n\mathbb{Z} \text{ is a field} \\ \Updownarrow \\ \mathbb{Z}/n\mathbb{Z} \text{ has no nonzero zero divisors} \\ \Updownarrow \\ n \text{ is prime.} \end{array}}$$

If $n = p$ is prime, every nonzero class has a representative $a \in \{1, \ldots, p-1\}$, so $\gcd(a, p) = 1$; the unit criterion says that every nonzero class is invertible.
If $n = rs$ is composite, with $1 < r, s < n$, then

$$\overline{r} \neq \overline{0}, \qquad \overline{s} \neq \overline{0}, \qquad \overline{r}\,\overline{s} = \overline{0}.$$

A field cannot contain such zero divisors.

# Prime modulus: $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{F}_p$

When $p$ is prime, the field

$$\mathbb{Z}/p\mathbb{Z} = \{\overline{0}, \overline{1}, \ldots, \overline{p-1}\}$$

is usually denoted by

$$\boxed{\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.}$$

It has $p$ elements and characteristic $p$:

$$p\,\overline{1} = \overline{0}.$$

## A notation warning

Some texts also write $\mathbb{Z}_p$ for the finite field $\mathbb{Z}/p\mathbb{Z}$. In these slides, $\mathbb{Z}_p$ already means the ring of *p-adic integers*: it is infinite and has characteristic 0. We therefore use $\mathbb{F}_p$ for the finite field.

These objects are closely related, but they are not equal:

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}, \qquad \mathbb{Z}_p = \{X \in \mathbb{Q}_p : \|X\|_p \leq 1\}.$$

# The group of units $(\mathbb{Z}/n\mathbb{Z})^\times$

Gather the invertible classes into a single set:

$$\boxed{(\mathbb{Z}/n\mathbb{Z})^\times = \{\overline{a} : \gcd(a, n) = 1\}.}$$

A product of units is a unit, $\overline{1}$ is a unit, and every unit has an inverse. Hence $(\mathbb{Z}/n\mathbb{Z})^\times$ is a finite abelian group under multiplication.

## Euler's totient

Its order is the number of residues coprime to $n$:

$$\varphi(n) = \#\{a : 1 \leq a \leq n,\ \gcd(a, n) = 1\}.$$

For a prime power the multiples of $p$ are the only residues to discard:

$$\varphi(p) = p - 1, \qquad \varphi(p^k) = p^k - p^{k-1}.$$

For example $\varphi(10) = 4$, matching the four units $\overline{1}, \overline{3}, \overline{7}, \overline{9}$.

# Fermat's little theorem and Euler's theorem

Recall Lagrange's theorem: in a finite group of order $N$, every element $x$ satisfies $x^N = e$. Applied to $(\mathbb{Z}/n\mathbb{Z})^\times$, whose order is $\varphi(n)$, it reads as follows.

## Euler's theorem

If $\gcd(a, n) = 1$, then

$$\boxed{a^{\varphi(n)} \equiv 1 \pmod{n}.}$$

For a prime modulus $n = p$ we have $\varphi(p) = p - 1$, giving the special case found by Fermat.

## Fermat's little theorem

For $p$ prime,

$$a^{p-1} \equiv 1 \pmod{p} \ \ (\gcd(a, p) = 1), \qquad a^p \equiv a \pmod{p} \ \ (\text{all } a).$$

For instance $3^4 = 81 \equiv 1 \pmod{10}$ and $2^{10} = 1024 \equiv 1 \pmod{11}$.

# The Chinese remainder theorem

When the moduli are coprime, arithmetic modulo a product splits into independent pieces.

**Theorem**

If $\gcd(m, n) = 1$, then reduction modulo $m$ and modulo $n$ gives a ring isomorphism

$$\boxed{\mathbb{Z}/mn\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \qquad \bar{a} \longmapsto (\bar{a}, \bar{a}).}$$

The map is a ring homomorphism. It is injective: if $m \mid a$ and $n \mid a$ with $\gcd(m, n) = 1$, then $mn \mid a$.
Both sides have $mn$ elements, so it is a bijection.
Equivalently, the simultaneous congruences

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

have a unique solution modulo $mn$. For example $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 5$ give $x \equiv 8 \pmod{15}$.

# Splitting into prime powers

Comparing units on both sides of the isomorphism,

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times \qquad (\gcd(m,n)=1),$$

so Euler's totient is multiplicative: $\varphi(mn)=\varphi(m)\varphi(n)$.
Iterating over the prime factorization $n = p_1^{e_1}\cdots p_k^{e_k}$ supplied by the fundamental theorem of arithmetic,

$$\boxed{\mathbb{Z}/n\mathbb{Z} \cong \prod_{i=1}^{k} \mathbb{Z}/p_i^{e_i}\mathbb{Z},} \qquad \varphi(n) = n\prod_{p\mid n}\left(1-\tfrac{1}{p}\right).$$

## A bridge to the $p$-adics

Every modulus thus dissolves into prime-power moduli $\mathbb{Z}/p^e\mathbb{Z}$. These are exactly the finite quotients of the $p$-adic integers $\mathbb{Z}_p$ — the objects of the next two slides.

# Reducing a $p$-adic integer modulo $p^m$

Recall that $X \in \mathbb{Z}_p$ has a unique canonical remainder $r_m \in \{0, \dots, p^m - 1\}$ at every level. Hence there is a surjective ring map

$$\mathrm{red}_m \colon \mathbb{Z}_p \longrightarrow \mathbb{Z}/p^m\mathbb{Z}, \qquad X \longmapsto \overline{r_m}.$$

Its kernel consists of the $p$-adic integers divisible by $p^m$:

$$\ker(\mathrm{red}_m) = p^m\mathbb{Z}_p.$$

The quotient therefore recovers the finite ring at level $m$:

$$\boxed{\mathbb{Z}_p/p^m\mathbb{Z}_p \cong \mathbb{Z}/p^m\mathbb{Z}.}$$

At the first level we obtain the residue field

$$\boxed{\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p.}$$

Thus $\mathbb{F}_p$ is a quotient of $\mathbb{Z}_p$, not a subring of it. Indeed, $\mathbb{Z}_p$ has characteristic 0, whereas $\mathbb{F}_p$ has characteristic $p$.

# All finite levels determine $\mathbb{Z}_p$

Reduction gives a tower of finite rings

$$\cdots \longrightarrow \mathbb{Z}/p^3\mathbb{Z} \longrightarrow \mathbb{Z}/p^2\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z},$$

where every arrow reduces a residue class to the preceding level.
A $p$-adic integer determines one class $x_m \in \mathbb{Z}/p^m\mathbb{Z}$ for every $m$, and the classes are compatible:

$$x_{m+1} \equiv x_m \pmod{p^m}.$$

Conversely, every such compatible family determines one $p$-adic integer. In compact notation,

$$\boxed{\mathbb{Z}_p \cong \varprojlim_m \mathbb{Z}/p^m\mathbb{Z}.}$$

## A useful contrast

The ring $\mathbb{Z}_p$ has no nonzero zero divisors because it sits inside the field $\mathbb{Q}_p$, but it is not a field: $p$ has no inverse in $\mathbb{Z}_p$. Its finite quotients $\mathbb{Z}/p^m\mathbb{Z}$ have zero divisors as soon as $m \geq 2$.

# Arithmetic modulo $n$

Fixing $n \geq 2$, we found:

- congruence $a \equiv b \pmod{n}$ is an equivalence relation, and its classes form the quotient ring $\mathbb{Z}/n\mathbb{Z}$, with all ring laws inherited from $\mathbb{Z}$;
- a class $\overline{a}$ is a unit exactly when $\gcd(a, n) = 1$, by Bézout, and a nonzero zero divisor otherwise;
- the units form a group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$, yielding Euler's theorem and, for a prime modulus, Fermat's little theorem;
- $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if $n$ is prime, giving the finite fields $\mathbb{F}_p$;
- the Chinese remainder theorem splits $\mathbb{Z}/n\mathbb{Z}$ into its prime-power parts $\mathbb{Z}/p^e\mathbb{Z}$;
- these prime-power rings are the finite quotients of $\mathbb{Z}_p$, and $\mathbb{Z}_p \cong \varprojlim_m \mathbb{Z}/p^m\mathbb{Z}$ reassembles them — closing the circle with the $p$-adic numbers.